

From Tripartite Synapses to Multicellular Engrams

Astrocytic State, Recall Precision, and Virtual Engram Stability

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Preprint; not peer reviewed. Synthetic and formal results are bounded to their declared models and do not establish a biological astrocyte memory role.

Abstract

A remembered object can remain recognizable even though the brain never returns to precisely the same cellular state. This creates a physical reconstruction problem: what must persist, what may change, and which cells control the content, availability, stability, or precision of recall? Neuronal-engram experiments establish distributed ensembles whose reactivation can contribute causally to recall. Astrocyte experiments add a second cellular layer: perisynaptic astrocytes detect transmitter and potassium conditions, transform local calcium and membrane state, regulate clearance and plasticity, and form learning- or recall-associated ensembles whose perturbation changes memory-related behavior. These findings make a multicellular account testable, but they do not by themselves distinguish content, index, gate, eligibility, stabilization, precision, motivation, affect, and general state. This paper develops a role-separated model of multicellular recall. A cue recruits a present neuronal ensemble inside a slower astrocytic, neuromodulatory, metabolic, and extracellular context. The model defines quantitative tests for content, access, stabilization, precision, and neuronal-astrocytic synergy; introduces a carrier-flexible virtual engram as a reconstructable relation that remains usable while its physical realization changes; and retains a Self-Aware Networks extension in which astrocytic state may alter the tonic reference against which receiving populations detect timing- and state-relative differences. The SAN term is introduced only after the operation is explained and remains an additional model to be tested against neuronal-only and global-state baselines. The paper specifies dual tagging, cross-perturbation, matched and mismatched rescue, neuronal bypass, recent-versus-remote and encoding-versus-recall contrasts, intervention-footprint assays, and complete-animal inference. Four executable synthetic programs then test the measurement logic. In the first, natural observations recover 12.5% of six role generators and falsely call 14.2% of non-content runs content; a selective intervention battery raises nominal recovery to 92.4%, but several shifts drive the model to unresolved decisions. An independently implemented dynamical generator reverses the ranking: observation reaches 97.5%, exceeding a 93.3% causal-endpoint model, while a capacity-matched temporal model reaches 35.0%. A 128-cell factorial envelope makes every pairwise model ranking reverse somewhere and shows that high conformal coverage can require diffuse two- or three-role sets. Continuous leave-one-generator-family-out transport then produces the strongest adverse result: models reaching 81.4%-93.3% in familiar regimes collapse to 2.4%-4.8% under an independently rotated measurement law, and target calibration can restore coverage only by returning nearly all six roles. A prospective multilevel simulator represents animals, sessions, laboratories, tools, promoters, memory age, intervention timing, specificity, footprint, and four target-information conditions. Its first allocation exposes a tool-promoter alias rather than hiding it. A separately frozen fully crossed repair places all nine tool-promoter cells in every laboratory and split, independently varies specificity and footprint, executes two label-free adapters and equal-capacity linear/nonlinear readouts, and retains 36,288

held-out predictions with exact replay. CORAL harms accuracy in three laboratories; empirical-quantile alignment does not repair transport; supervised target refitting improves accuracy but is not label-free transport; and none of 224 factorial readbacks survives its global Holm family. Assumption-conditional power calculations show that structural identifiability does not guarantee adequate precision. A bounded Lean 4 certificate checks twenty-six finite allocation, count, access, capacity, evidence-history, interpretation, scope, and seal invariants with no axiom dependencies. It does not import the CSV files, verify the Python implementation, prove biological causal assumptions, identify an astrocyte role, or validate SAN, PWD, NAPOT, consciousness, treatment, safety, or a clinical claim. The biological endpoint remains sealed. The contribution is therefore a source-faithful theory-and-methods architecture, an executable record of favorable and adverse identifiability regimes, and a concrete path by which a future experiment could distinguish multicellular memory roles without converting correlation, perturbation success, coverage, or formal bookkeeping into content language.

Keywords: astrocyte; engram; memory recall; tripartite synapse; representational drift; virtual engram; Self-Aware Networks; NAPOT; phase-wave differential; neural reconstruction

1. The ordinary problem: the remembered broccoli is never physically identical

You can recognize broccoli under different lighting, from a new angle, after years of bodily and neural change, and while your current attention and emotional state differ from the first encounter. The perceptual and mnemonic result is stable enough to use, yet the physical state producing it is not a stored photograph replayed without alteration. Neurons change their excitability and synaptic partners. Molecular and metabolic conditions change. Population representations can drift across days or weeks while behavior remains stable [8,9].

This is the problem before any new terminology is introduced:

How can a changing biological network reconstruct a usable relation to a past experience without requiring the exact same cells, voltages, molecular states, and timing pattern to recur?

One answer is that a subset of neurons changed during learning and can later be reactivated. This neuronal-engram account is experimentally powerful. Learning-activated neurons can be tagged, manipulated, and connected across brain regions; a single contextual fear memory recruits a distributed complex rather than one isolated site [7]. But a neuronal label does not measure every state variable that influences what those neurons do. A synapse is embedded in extracellular chemistry, neuromodulation, metabolism, vasculature, microglia, oligodendrocytes, and astrocyte processes. The current neuronal pattern therefore operates within a cellular environment partly created by other cells.

The next question is not whether neurons or astrocytes “own” memory. It is which physical operations each state variable performs during learning and recall.

2. What established neuroscience already explains

2.1 Distributed neuronal engrams

An engram is commonly operationalized through lasting physical or chemical change induced by learning, selective reactivation during recall, and causal effects of manipulating the relevant cells. Modern experiments support neuronal engrams distributed across hippocampal, amygdalar, cortical, thalamic, hypothalamic, and brainstem regions. In contextual fear conditioning, Roy and colleagues mapped candidate ensembles across 247 mouse brain regions and found that simultaneous activation of multiple confirmed ensembles produced stronger recall than activation of one ensemble [7]. This establishes an important baseline: recall is already a network reconstruction problem, not a one-cell or one-region replay.

Longitudinal recording adds a second baseline. Stimulus representations can change across sessions in visual and olfactory cortex even when the animal continues to discriminate or use the stimulus [8,9]. Drift does not prove that every memory uses changing carriers, but it prevents a theory from equating stable function with byte-for-byte recurrence of one population state.

2.2 The local astrocyte–synapse operation

Astrocytes are positioned to transform the conditions under which synapses work. Fine processes approach many synapses and express receptors, channels, transporters, and intracellular signaling machinery. In adult mouse hippocampus, local astrocyte calcium events detected nearby synaptic activity, while disrupting those signals reduced transmission reliability under minimal stimulation [1]. Other experiments linked astrocyte state to D-serine-dependent LTP [2], perisynaptic coverage and glutamate spillover [3], and potassium-clearance contributions to short-term plasticity [4].

Voltage measurements added a complementary operation. Armbruster and colleagues observed rapid, focal, pathway-specific depolarization in peripheral astrocyte processes during neuronal activity. Extracellular potassium and electrogenic glutamate transport contributed to the response, and depolarization slowed glutamate clearance in the tested preparation [6]. The important operation is concrete: a local neuronal event changes astrocyte state; the changed astrocyte state alters the extracellular and synaptic conditions encountered by subsequent neuronal activity.

Only after this operation is visible is the inherited term **tripartite synapse** useful. It names the functional participation of presynaptic terminal, postsynaptic target, and astrocytic process. It does not imply that all synapses share one astrocyte mechanism or that a local astrocyte process contains a complete remembered scene.

2.3 Learning-associated astrocyte state

Astrocyte effects extend beyond momentary synaptic regulation. Gq activation of CA1 astrocytes induced persistent neuronal potentiation and altered memory acquisition and allocation in mice [5]. Rupprecht and colleagues found that hippocampal astrocyte activity integrated past neuronal and behavioral events over seconds, with salient events and locus-coeruleus signaling facilitating process-to-soma propagation [10]. This result is especially informative because it reveals both computational capacity and a confound: astrocyte activity can reflect arousal and neuromodulatory state as well as local neuronal events.

Recent ensemble studies make direct role tests possible. Spatial transcriptomics identified persistent, memory-associated neuronal and astrocytic subpopulations in the mouse basolateral amygdala and supported functional interaction between adjacent astrocytes and neurons during long-term consolidation [11]. Williamson and colleagues identified learning-associated hippocampal astrocyte ensembles affiliated with neuronal engrams; astrocyte-ensemble reactivation elicited recall, while ensemble-specific NFIA deletion impaired it [12]. Serra and colleagues used AstroLight to manipulate task-recruited nucleus-accumbens astrocyte ensembles and changed cue-motivated behavior [13]. Dewa and colleagues identified an astrocyte ensemble that was primed by an initial fear experience and recruited by coincident local-engram and noradrenergic signals during later recall, affecting memory stability and precision over days [14]. Bukalo and colleagues showed that basolateral-amygdala astrocyte calcium tracked fear state and supported neuronal conditioned-stimulus representations, cortical readout, retrieval, and extinction [15]. Cooper and colleagues further mapped reproducible, region-specific astrocyte networks coupled through gap junctions, expanding the anatomical scope of the glial network question without by itself establishing a memory code or its bandwidth [16].

These experiments do not all measure the same thing. Recall, consolidation, precision, motivated approach, fear state, and neuronal representation are distinct endpoints. Their diversity is not a weakness. It supplies the experimental leverage needed to distinguish causal roles.

3. The unresolved role-identification problem

Suppose reactivating a learning-tagged astrocyte ensemble causes freezing. At least six explanations remain possible:

1. **Content:** the astrocyte state carries information that discriminates the remembered cue from alternatives.
2. **Index:** it points the network toward the neuronal ensemble that carries discriminating content.
3. **Gate:** it changes whether an otherwise available neuronal representation can control downstream behavior.
4. **Eligibility trace:** an earlier state marks a local circuit so that a later event produces stronger plasticity or stabilization.
5. **Stabilizer:** it changes how long or how precisely the memory persists without specifying its identity.
6. **General state:** it changes fear, arousal, locomotion, motivation, or excitability, thereby changing the behavioral readout without reconstructing the remembered content.

The phrase “astrocytes are part of the engram” is therefore the beginning of an experiment, not the end of the explanation. A useful theory must state what would count as each role and what result would distinguish it from the others.

4. A source-faithful SAN genealogy

The present proposal develops from a longer Self-Aware Networks research line, but later terminology should not be projected backward.

4.1 The 2017 public architecture

In the fourth public Neural Lace Podcast, I considered the limits of a hypothetical thirty-second “snapshot” of every neuron and astrocyte. The discussion asked how a neuron-only recording could predict change under differences in testosterone, psychedelic state, or slower receptor-mediated chemistry. It described a neuronal network and a “glionet” interfacing across fast ionotropic and slower metabotropic processes. The conversation then moved to a reconstruction problem: no exact brain signal may recur, yet the brain recognizes the same headphones, bottle, glass, or broccoli across new angles, lighting, rooms, and moments. A future read/write system would therefore need to decode relations that remain usable across changing physical patterns and manage the conflict between an inserted representation and current sensory context [17].

The complete argument matters. Reduced to “astrocytes matter,” it loses the fast/slow state distinction, the inadequacy of an instantaneous neuron-only snapshot, the changing-carrier problem, and the reconstruction requirement. The 2017 source is a public ancestor of the present multicellular-state model. It does not contain the later astrocyte-ensemble results or the terms COT, NAPOT, PWD, virtual engram, or astroengram.

4.2 The 2022 public mechanism

Public SAN Git notes in 2022 made the architecture more explicit. One source joined multimodal reconstruction, array-to-array projection, phase-coincidence relations, synchronized and splay states, distributed receiving populations, and structural change in neurons and glia [18]. Another developed receptor configuration, cellular thresholds, morphology, coincidence-sensitive response, and learned cell state, then scaled this proposal through Cellular Oscillating Tomography (COT) toward neuronal and glial participation in Neural Array Projection Oscillation Tomography (NAPOT) [19].

The strongest faithful reading is not “every cell stores a whole memory.” It is that a cell’s learned morphology, receptor composition, intracellular state, and network position change what patterns it detects, how it transforms them, and what it transmits next. At the network level, usable content is reconstructed through relations among changing receivers and transmitters.

4.3 What is new in the present paper

The 2024–2026 astrocyte-ensemble literature provides causal handles that the earlier architecture lacked. The present paper adds a formal role taxonomy, identifiability criteria, a cross-perturbation design, and an explicit sequence of nested models. This joined multicellular-engram formulation is a 2026 synthesis. Earlier sources establish its genealogy but do not inherit its later equations or molecular details.

5. The proposed operation in plain language

A cue first activates sensory and associative pathways. The recruited neurons do not operate in an empty space. Nearby astrocytes have a current membrane, calcium, transporter, metabolic, receptor, and transcriptional state shaped by earlier activity and neuromodulation. That state can change transmitter clearance, extracellular potassium, synaptic gain, plasticity thresholds, and which local pathways remain effective.

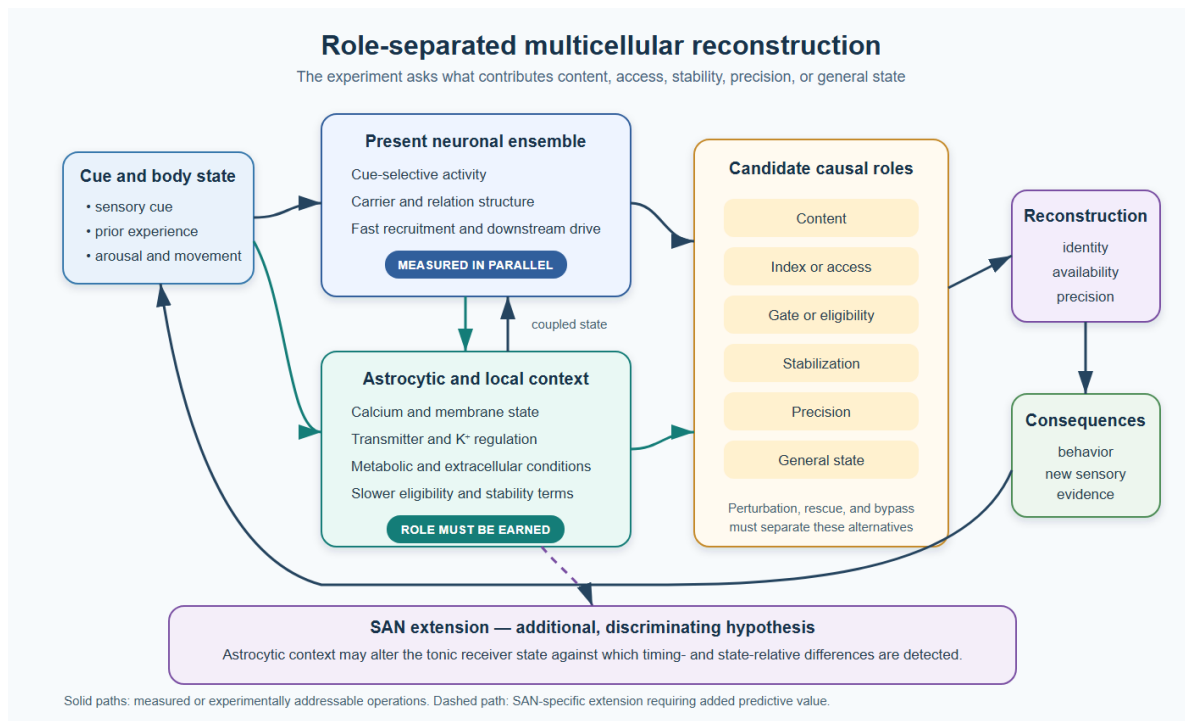
During learning, neuronal and astrocytic states may both change. During later recall, the cue recruits a present neuronal ensemble inside that altered local context. The astrocyte state may help open a route, bias which ensemble wins, stabilize a labile trace, sharpen or broaden generalization, or participate in a distributed pattern that discriminates one cue from another. The reconstructed memory is the usable downstream relation produced by the current system, not necessarily a frozen list of original cells.

This yields the central causal chain:

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cue and current body state
-> neuronal candidate reconstruction
-> local astrocyte and neuromodulatory context
-> changed clearance, gain, route eligibility, and plasticity
-> selected multicellular trajectory
-> cue-specific output, access, stability, and precision
-> new action and sensory feedback

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After the operation is understood, two new labels become useful.

A **multicellular engram** is a memory-related physical trace whose causal operation is distributed across more than one cell class, with the role of each class stated rather than assumed.

A **virtual engram** is the carrier-flexible relation reconstructed by the current system when the same usable memory remains identifiable despite change in the particular cells or activity pattern realizing it.

6. Formal model

6.1 State and dynamics

Let the measured system state at time (t) be

$$S_t = \{N_t, A_t, U_t, X_t\}, \quad (1)$$

where N_t is the neuronal population state, A_t is the astrocyte state, U_t is neuromodulatory and extracellular context, and X_t contains observed behavioral and physiological covariates such as locomotion, pupil diameter, respiration, shock history, reward expectation, and current cue.

The coupled dynamics are

$$N_{t+1} = F_N(N_t, A_t, U_t, C_t; \theta_N) + \varepsilon_t^N, \quad (2)$$

$$A_{t+1} = F_A(A_t, N_t, U_t, C_t; \theta_A, \tau_A) + \varepsilon_t^A, \quad (3)$$

where C_t is the cue, τ_A allows astrocytic variables to integrate over different timescales, and the noise terms include unmeasured state. Equations (2) and (3) do not presume that astrocytes are uniformly slower than neurons. They permit local voltage and calcium effects on fast timescales alongside transcriptional and eligibility effects lasting seconds to days.

Recall is a measured output rather than a binary metaphysical state:

$$R_t = G(N_t, A_t, U_t, C_t, X_t), \quad (4)$$

with separate components for cue identity, access probability, latency, behavioral policy, confidence or generalization, and persistence.

6.2 Role scores

Let Y denote the learned cue or episode identity. Astrocytic content requires discriminating information beyond the measured neuronal and global-state variables:

$$\mathcal{C}_A = I(Y; A_t \mid N_t, C_t, U_t, X_t), \quad (5)$$

where I is conditional mutual information. A positive estimate is insufficient by itself because hidden neuronal variables can create residual association. A content disposition therefore also requires selective causal manipulation and cue-specific rescue.

An access or gate effect is

$$\mathcal{G}_A = P(R_{access} = 1 \mid do(A = A^{on})) - P(R_{access} = 1 \mid do(A = A^{off})), \quad (6)$$

evaluated while cue identity, gross neuronal decodability, and general-state covariates are matched as closely as feasible.

A stabilization effect over delay d is

$$\mathcal{S}_A(d) = \left[-\frac{\partial}{\partial d} P(R_{correct} = 1 \mid d) \right]_{A^{off}} - \left[-\frac{\partial}{\partial d} P(R_{correct} = 1 \mid d) \right]_{A^{on}}, \quad (7)$$

so that a positive value means the declared astrocyte intervention reduces the rate of performance loss across the specified delay. The sign and intervention direction must be preregistered; the essential distinction is a change in the decay or restabilization trajectory rather than only immediate expression.

Precision is separated from access:

$$\mathcal{P}_A = \Delta[P(R = Y \mid C = Y) - P(R = Y \mid C \neq Y)]_{do(A)}. \quad (8)$$

This contrast distinguishes selective recall from generalized freezing, approach, or arousal.

6.3 Changing carriers and reconstructable invariance

Let E_t be the set of neurons strongly associated with the memory at time t . Its longitudinal overlap can be written as a Jaccard index:

$$J_N(t, t + d) = \frac{|E_t \cap E_{t+d}|}{|E_t \cup E_{t+d}|}. \quad (9)$$

A carrier-flexible reconstruction occurs when neuronal overlap decreases while the task-relevant relation remains recoverable:

$$J_N(t, t + d) < \eta \quad \wedge \quad I(Y; R_{t+d}) \geq \kappa, \quad (10)$$

for preregistered thresholds η and κ . Equation (10) earns the term **virtual engram** only at the operational level: the remembered relation is recoverable across changing realization. It does not imply that the memory is nonphysical or substrate-independent.

6.4 Neuronal–astrocytic synergy

The simplest causal synergy statistic for a factorial perturbation is

$$\mathcal{I}_{NA} = R_{11} - R_{10} - R_{01} + R_{00}, \quad (11)$$

where the indices denote neuronal-ensemble and astrocyte-ensemble availability. A nonzero interaction says that the joint effect is not the additive sum of independent manipulations. It does not identify the role without the content, access, stability, and precision endpoints above.

6.5 Identifiability and hidden-state equivalence

The same behavioral endpoint can be generated by different causal roles. The role-identification analysis therefore treats role identification as a model-comparison problem, not as an interpretation

of one successful manipulation. Let (M_r) denote a role model, (O) the observed recordings and covariates, and $(\setminus I)$ the available intervention set. Two role models are observationally equivalent over an experiment when

$$P_{M_r}(O \mid C) = P_{M_s}(O \mid C) \quad \text{for every measured cue and history } C. \quad (12)$$

Observational equivalence is broken only when an admissible intervention produces a distinguishable distribution or a preregistered rescue effect:

$$\exists I \in \mathcal{I} : P_{M_r}(O \mid do(I), C) \neq P_{M_s}(O \mid do(I), C). \quad (13)$$

This formalizes why astrocyte decoding alone cannot establish content. A hidden neuronal state (H_N) can carry the discriminating information while remaining absent from the recorded feature set. The content estimator must therefore be stress-tested under latent-state surrogates:

$$\hat{\mathcal{C}}_A^{\text{resid}} = I(Y; A \mid N_{\text{obs}}, U, X, H_N^*), \quad (14)$$

where (H_N^*) is a family of held-out latent-neuronal reconstructions generated without access to the target label. A positive residual is evidence for a candidate role only if it survives that family and the selective causal tests.

For a role claim to be identifiable in the declared experiment, at least one intervention must separate it from every retained alternative while preserving cue, gross state, and tool-footprint controls:

$$\text{ID}(M_r) = 1 \iff \forall M_s \neq M_r \exists I \in \mathcal{I} : \Delta_{r,s}(I) > \delta_{r,s}. \quad (15)$$

If no such intervention exists, the correct result is an equivalence class of roles, not a forced winner. This is a mathematical reason to report unresolved identifiability regions rather than overinterpret a positive behavioral effect.

7. The SAN receiver-relative extension

The established model already contains a current background state and event-driven departures from it. SAN adds a more specific possibility. A receiving population is not a passive wire. Its membrane state, inhibition, recent activity, learned morphology, astrocyte environment, and expected timing determine what an arriving event does. The informative variable may therefore be the difference between an arriving event and the relation expected by that receiver under its current context.

For sender i and receiver j , define

$$D_{j \leftarrow i}(t) = \text{wrap}[\phi_i(t - \tau_{ij}) - \phi_j(t) - \mu_{ij}(S_t)], \quad (16)$$

where ϕ denotes a measurable phase estimate, τ_{ij} is propagation delay, and $\mu_{ij}(S_t)$ is the expected relation conditioned on the current multicellular state. SAN calls a consequential receiver-relative departure of this kind a **Phase-Wave Differential** (PWD). The term is narrower than phase coding or a traveling wave: it specifies sender, receiver, expected relation, delay, measured deviation, and downstream consequence.

In this paper, astrocyte state is proposed to alter $\mu_{ij}(S_t)$, the gain applied to $D_{j \leftarrow i}(t)$, or the persistence of the selected route. The added claim survives only if receiver-relative variables improve held-out prediction or selective causal control beyond models using neuronal rate, ordinary phase, astrocyte calcium, arousal, locomotion, cell identity, and global network state.

8. Four nested models

The experiment should compare nested explanations rather than test SAN against no model.

Model 0: neuronal engram only

Recall is predicted from current neuronal ensemble state, cue, and measured behavioral covariates. Astrocyte variables add no reliable held-out value after controls.

Model 1: global astrocyte state

Astrocyte activity improves prediction because it tracks arousal, fear, motivation, or general circuit excitability. Spatially selected learning-associated astrocytes are unnecessary once global state is measured.

Model 2: role-separated multicellular engram

Selected astrocyte state contributes to one or more specified roles—index, gate, eligibility, stabilization, precision, or content—and cross-perturbation reveals a reproducible role-specific causal effect.

Model 3: SAN receiver-relative reconstruction

Model 2 is extended with receiver-relative timing and state differences. It is favored only if these terms improve preregistered held-out prediction, survive state-matched controls, and support a selective perturbation/rescue result.

Comparison should use held-out log likelihood, calibrated classification, effect-size uncertainty, and predeclared complexity penalties rather than training fit alone.

9. Decisive experiment

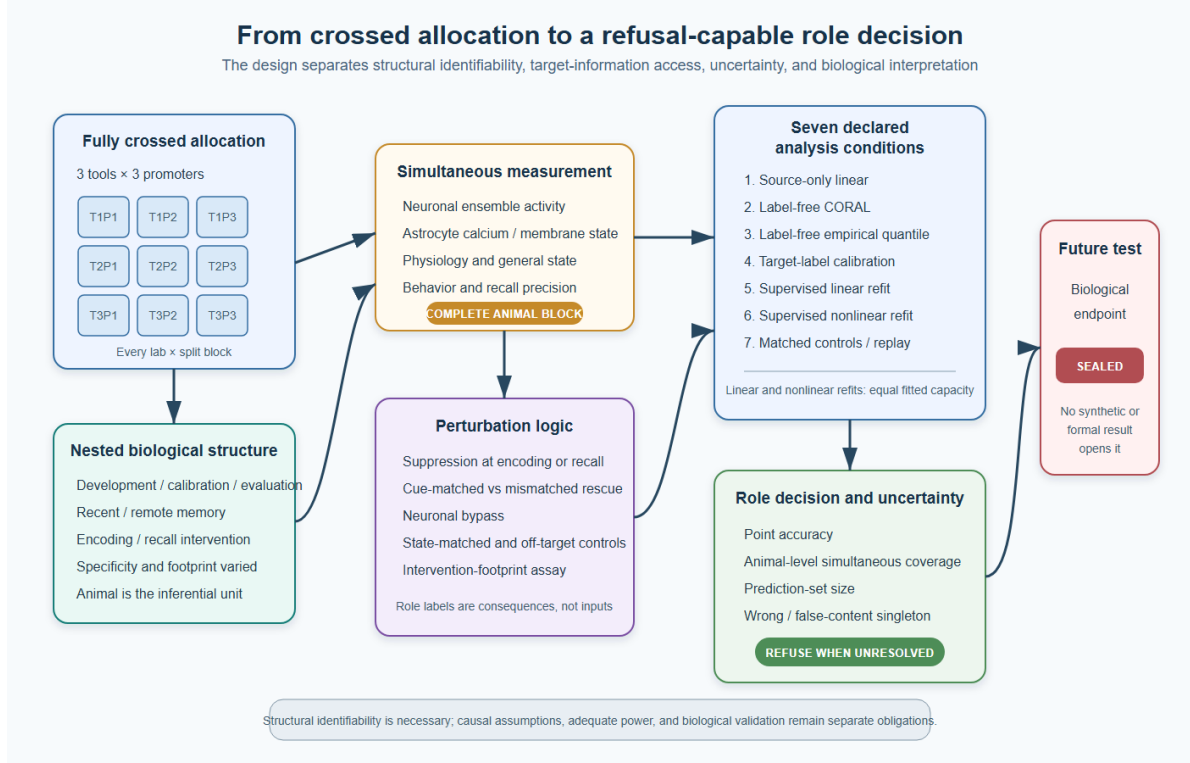
9.1 Design

Use two discriminable cues, Y_1 and Y_2 , with matched sensory salience but different learned outcomes. Tag learning-associated neuronal and astrocyte ensembles for each cue using independently addressable systems. Measure neuronal activity, astrocyte calcium and—where feasible—astrocyte membrane voltage, noradrenergic state, pupil diameter, locomotion, respiration, and behavior during encoding, natural recall, extinction, and delayed recall.

The core factorial conditions are:

1. both tagged ensembles available;
2. neuronal ensemble suppressed, astrocyte ensemble available;
3. astrocyte ensemble suppressed, neuronal ensemble available;
4. both suppressed;
5. astrocyte ensemble reactivated without its cue;
6. neuronal ensemble reactivated without its cue;
7. mechanism-matched rescue after each selective suppression;
8. off-target and untagged-cell controls matched for light, ligand, virus, handling, and behavioral history.

Repeat the same memory at multiple delays to estimate immediate access, restabilization, long-term retention, precision, and changing ensemble overlap.



9.2 Content test

The strongest astrocyte-content result would satisfy all of the following:

- astrocyte state decodes Y_1 versus Y_2 after measured neuronal and global-state variables are conditioned out;
- selective astrocyte reactivation produces cue-appropriate discrimination, not merely increased freezing or approach;
- the result persists when the associated neuronal engram is suppressed strongly enough to reduce its measured decodability;

- a cue-matched astrocyte rescue restores the correct memory more than an equally strong mismatched rescue;
- affect, arousal, locomotion, motivation, and intervention spread cannot explain the difference.

Failure of this stringent test would not make astrocytes irrelevant. It would move the supported role toward indexing, gating, eligibility, stabilization, precision, or global state.

9.3 Stabilization and eligibility test

Dewa and colleagues motivate a two-stage design [14]. Manipulate the initial experience-dependent astrocyte state separately from the recall-recruited state. Measure whether the manipulation affects immediate recall, post-recall lability, later retention, and generalization. A selective effect on later stability with intact initial recall supports an eligibility/stabilization role more directly than a content role.

9.4 Tool-footprint controls

Astrocyte tools can have slow kinetics, incomplete selectivity, or effects that spread through coupled networks. Each causal interpretation therefore requires:

- histological cell-type specificity;
- measured onset and offset kinetics;
- three-dimensional spatial footprint;
- dose-response relation;
- effects on neighboring untagged astrocytes and neurons;
- gap-junction and extracellular-spread controls where appropriate;
- sham and effector-only controls;
- analysis blinded to condition;
- animal-level inference rather than treating cells as independent animals.

9.5 General-state controls

Fear-memory studies must distinguish remembered cue identity from fear expression. Reward studies must distinguish learned cue identity from general motivation. At minimum, measure open-field activity, pupil state, respiration, shock reactivity, reward consumption, baseline preference, anxiety-related behavior, and cue discrimination. The relevant question is not simply whether behavior changes, but which component of recall changes after the intervention.

9.6 Measurement and timing contract

Variable	Instrument or assay	Minimum timing role	Primary analysis endpoint
Neuronal ensemble activity	Cell-type-resolved calcium imaging or electrophysiology	Encoding, immediate recall, delayed recall	Cue decoding, ensemble overlap, neuronal suppression check
Astrocyte calcium	Astrocyte-targeted calcium imaging	Process-to-soma latency and recall epochs	Role-specific activity, propagation, cue/state decoding
Astrocyte membrane state	Genetically encoded voltage indicator where validated	Fast local response around synaptic events	Local depolarization and receiver-relative state term
Noradrenergic context	Fluorescent sensor or locus-coeruleus activity proxy	Encoding and recall modulation	Arousal/eligibility covariate and interaction
General state	Pupil, respiration, locomotion, shock/reward reactivity	Continuous or trial-locked	State-matched controls and nuisance regression
Recall behavior	Cue discrimination, freezing/approach, latency, generalization	Immediate and delayed endpoints	Content, access, stability, precision, and motivation separated
Tool footprint	Histology, light/ligand timing, spatial spread, off-target assays	Before and after causal runs	Exclusion, selectivity, and dose-response certification

Animal is the inferential unit for causal effects. Cells and trials provide repeated measurements nested inside animals; they do not create independent animals. The primary analysis must be sealed before unblinding and must report missingness, exclusions, intervention failures, and the full timing of each perturbation.

10. Executable role-identifiability benchmark

10.1 The problem the application must solve

An observational decoder can appear to find an astrocyte memory code even when astrocytes only index neurons, open access, preserve a trace, stabilize recall, or track general state. The first frozen benchmark therefore does not ask whether a flexible model can fit simulated behavior. It asks whether three predeclared decision systems recover the correct causal role when all six generators look similar during natural recall.

The benchmark contains content, index, gate, eligibility, stabilizer, and global-state generators. Each produces cue-correlated neuronal and astrocyte observations and five measured endpoints: identity, access, neuronal recruitment, delayed recall, and precision. Roles become separable only through nine conditions: natural recall; astrocyte suppression at recall; cue-matched and cue-mismatched astrocyte rescue; neuronal rescue after astrocyte suppression; encoding-only astrocyte suppression; post-recall suppression; global-state-matched suppression; and an off-target control.

10.2 Frozen design

The contract and executable source were hashed before evaluation. Forty-eight nominal development seeds per role define standardized centroids and an unresolved-margin threshold. Twenty-four disjoint evaluation seeds per role are then tested under nominal observation, hidden-neuronal confounding, partial observation, intervention spread, animal-level autocorrelation, state-measurement noise, and combined shift. This yields 288 development experiments and 1,008 held-out experiments. Each experiment contains nested animal and trial measurements; the experiment seed, not a cell or trial, is the independent synthetic unit.

Three classifiers use the same nearest-centroid decision rule:

1. **Observation only:** eight natural-recall features.
2. **Cross-perturbation:** the natural features plus fifteen intervention and rescue contrasts.
3. **Cross-perturbation plus latent surrogate:** the same causal features plus an astrocyte residual adjusted by a noisy, label-blind reconstruction of latent neuronal state.

For a frozen feature vector x , standardization uses development-only moments,

$$z_k(x) = \frac{x_k - \hat{\mu}_k^{dev}}{\hat{\sigma}_k^{dev}}, \quad (17)$$

and each synthetic role r has the development centroid

$$c_r = \frac{1}{|D_r|} \sum_{x \in D_r} z(x). \quad (18)$$

Let $d_{(1)}$ and $d_{(2)}$ be the smallest and second-smallest Euclidean distances from $z(x)$ to the six centroids. The classifier returns the nearest role only when its separation margin clears the development-frozen threshold $q_{0.05}$:

$$\hat{r}(x) = \begin{cases} \arg \min_r \|z(x) - c_r\|_2, & d_{(2)} - d_{(1)} \geq q_{0.05}, \\ \text{unresolved}, & d_{(2)} - d_{(1)} < q_{0.05}. \end{cases} \quad (19)$$

For every classifier, the unresolved threshold is the fifth percentile of the margin among correctly classified development experiments. The primary endpoints are role accuracy, unresolved rate, and false-content rate among non-content generators. Wilson 95% intervals are reported across held-out experiment seeds. No model is forced to choose a role when the frozen evidence falls outside development-supported separation.

10.3 Nominal results

Natural observations do not identify role. The observation-only classifier correctly labels 18 of 144 nominal experiments, an accuracy of 0.125 (95% Wilson interval 0.081-0.189), and leaves 0.132 unresolved. More importantly, it calls 17 of 120 non-content experiments content, a false-content rate of 0.142. Cue-correlated astrocyte activity is therefore compatible with several causal stories in this benchmark.

Cross-perturbation changes the result. It correctly labels 133 of 144 nominal experiments, or 0.924 (95% interval 0.868-0.957), leaving 0.076 unresolved. Adding the noisy latent-neuronal surrogate yields 135 of 144 correct, or 0.938 (95% interval 0.885-0.967), with 0.063 unresolved. Neither causal classifier falsely labels a nominal non-content generator as content. The remaining nominal errors are refusals: no wrong-role answer replaces uncertainty.

10.4 Adverse and unresolved results

The nominal result is not a general role detector. The causal classifier becomes completely unresolved under hidden-neuronal confounding, partial observation, autocorrelation, and combined shift. Its accuracy is therefore zero in those conditions because a refusal is not counted as a correct answer. State-measurement noise produces 0.097 accuracy and 0.903 unresolved decisions. Intervention spread is the most revealing non-refusal failure: accuracy falls to 0.167, and unresolved decisions rise to 0.333 because nonspecific perturbation erases the role-specific causal signature. The latent-surrogate model does not repair these shifts; under several of them it also refuses every experiment.

The observation-only classifier remains near chance across every adverse condition, with accuracy from 0.139 to 0.201. Its nominal false-content result is not repeated uniformly under all shifts, which is another warning: one confound can manufacture an apparent content result while a different shift can hide it. A single decoding score cannot certify the role.

10.5 Interpretation and exact boundary

The benchmark establishes three finite implementation results. First, the role definitions are observationally confusable by construction. Second, the declared cross-perturbation battery separates those roles under the nominal generator and measurement contract. Third, refusal exposes when that learned separation no longer transports. Exact replay reproduces all seven output-file hashes, and 26 deterministic contract and result tests pass.

The benchmark does not show that a biological astrocyte ensemble performs any one role. Its noisy latent surrogate is not a complete measurement of hidden neuronal state. Its effect sizes are transparent synthetic fixtures chosen to test identifiability, not parameters fitted to one published animal experiment. A second generator must therefore test whether the result survives different dynamics and carrier change. The biological test remains sealed.

10.6 Independent dynamical generator

The independent dynamical benchmark implements that second test with a separately coded generator. Each experiment contains sixteen time steps divided into encoding, recall, and post-recall epochs. It evolves astrocyte and neuronal state according to a declared state-space family,

$$A_{t+1} = \rho_A A_t + B_A(C, H, G, I, t) + \epsilon_t^A, \quad (20)$$

$$N_{t+1} = \rho_N N_t + B_N(C, H, A_t, I, r) + \epsilon_t^N, \quad (21)$$

where C is cue, H is hidden neuronal state, G is global state, I is the stage-specific intervention, and r is the generating role. The functions B_A and B_N are transparent fixtures, not fitted physiological

equations. Content, index, gate, eligibility, stabilizer, and global-state roles differ in which causal path changes output. Natural cue correlation is present in every role, but unlike the first frozen benchmark, the role-specific path can alter natural dynamics enough to become observable.

The neuronal vector uses a twelve-dimensional carrier that rotates between an early and late direction:

$$v_t = \frac{(1 - \alpha_t)v_0 + \alpha_tv_1}{\|(1 - \alpha_t)v_0 + \alpha_tv_1\|_2}, \quad \alpha_t = \frac{t}{T - 1}. \quad (22)$$

The study compares an early fixed-address decoder with a relation-preserving alignment. The latter estimates the orthogonal Procrustes map from unlabeled early and late activity,

$$Q^* = \arg \min_{Q^\top Q = I} \|X_{late}Q - X_{early}\|_F^2, \quad (23)$$

then applies the early cue decoder after alignment. This is a bounded synthetic test of carrier-flexible recovery. It is not evidence that a brain performs Procrustes alignment or that a biological virtual engram exists.

10.7 Capacity-matched classifier contract

Three primary role classifiers receive different sixteen-feature views: natural observations, causal endpoint contrasts, or temporal dynamics. Each has an intercept, sixteen standardized inputs, six outputs, and 102 fitted coefficients. A fourth control uses the temporal architecture with shuffled development labels. All use the same ridge solution,

$$\widehat{W} = (X^\top X + \lambda P)^{-1} X^\top Y, \quad (24)$$

where the intercept is unpenalized. Thirty-six nominal development seeds per role and twenty disjoint evaluation seeds per role and condition produce 216 development experiments and 840 held-out experiments. The seven evaluation conditions are nominal dynamics, carrier drift, hidden neuronal loading, intervention spread, partial observation, autocorrelation, and combined shift. The same development-only margin rule permits an unresolved answer.

10.8 Role-recovery reversal

The independent generator does not reproduce the first frozen benchmark’s observational confusability. The observation ridge correctly identifies 117 of 120 nominal experiments, or 0.975 (95% Wilson interval 0.929-0.991), and leaves 0.025 unresolved. The causal-endpoint ridge identifies 112 of 120, or 0.933 (95% interval 0.874-0.966), with 0.067 unresolved. The capacity-matched temporal model identifies only 42 of 120, or 0.350 (95% interval 0.271-0.439), and falsely calls 8.0% of non-content runs content. The shuffled-label control reaches 16 of 120, or 0.133.

This reversal narrows the earlier conclusion. Natural observation can identify role when the generating roles leave separable natural-state signatures. Cross-perturbation is not automatically more accurate. Its value in this generator is transport: under carrier drift, the causal-endpoint model remains at 0.900 accuracy while the observation model falls to 0.492. The causal model

reaches 0.558 under hidden neuronal loading, 0.717 under intervention spread, 0.575 under partial observation, 0.783 under autocorrelation, and 0.508 under combined shift. Every value remains synthetic and several are close enough to failure that a biological claim would be unjustified.

10.9 Carrier-change result and virtual-engram boundary

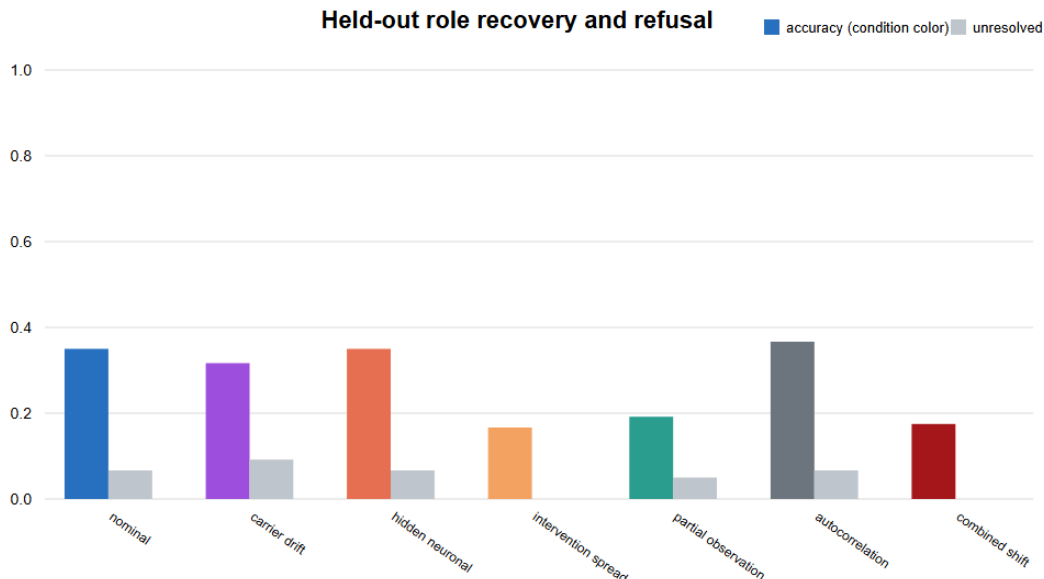
The carrier-drift condition reduces mean early-to-late carrier cosine to 0.087 and mean top-four address Jaccard overlap to 0.194. Nevertheless, the fixed-address cue decoder remains at 1.000 accuracy, exactly matching relational alignment. Carrier rotation alone therefore does not establish that a relational method is needed. Under combined shift, fixed-address accuracy averages 0.947 and alignment 0.999, a mean gain of 0.052 with large between-run variation. That is a bounded recovery effect, not a virtual-engram verdict.

The null matters. A low carrier cosine can coexist with preserved projection onto the early decoder, and high task accuracy can survive large address changes. The factorial envelope must generate carrier changes that independently vary overlap, fixed-decoder projection, relational invariance, and task performance. Only then can the paper test which invariant is necessary rather than merely display drift.

10.10 Ablations, figure, and exact replay

The temporal model’s low nominal result is not rescued by removing registered features. Removing recall-suppression features lowers accuracy from 0.350 to 0.283; removing encoding timing lowers it to 0.267; removing the receiver residual lowers it to 0.217. Removing post-recall timing has little effect at 0.333. Removing matched-versus-mismatched rescue slightly improves accuracy to 0.375, an adverse reversal that blocks a unique-necessity claim for that feature in this model.

Figure 3 reports accuracy and unresolved rate for the temporal-dynamics ridge only; it is not a summary of the stronger observation or causal-endpoint models. The SVG and its inspected PNG render are retained separately. A full second run reproduces every result and figure hash exactly. Twelve contract tests and fifteen result tests pass. The biological test remains sealed.



10.11 Why a factorial envelope is necessary

The first benchmark showed a regime in which natural cue correlation was misleading and selective intervention recovered the generating role. The independent benchmark then showed the opposite: when the six roles left separable natural signatures, observation was the strongest nominal classifier, although causal endpoints transported better under several shifts. Those results cannot be reconciled by choosing the preferred demonstration. They expose the missing problem: a role-identification method is useful only if we know the measurement conditions under which it succeeds, refuses, or reverses.

The factorial envelope therefore does not add a third isolated success case. It varies seven measurement and system properties independently at low and high levels: observational separability, causal separability, carrier-address overlap, preservation of the fixed decoder’s task projection, latent-neuronal loading, intervention spread, and measurement noise. The complete design is

$$\mathcal{E} = \{L, H\}^7, \quad |\mathcal{E}| = 2^7 = 128. \quad (25)$$

Each cell contains the same six candidate roles. Twelve animals contribute to every generated experiment. The generator produces observed-state summaries, causal summaries, between-animal moments, sixteen-step four-channel sequences, and independently controlled carrier-address and task-direction measures. Development seeds 5000-5005, calibration seeds 5100-5111, and evaluation seeds 6000-6011 are disjoint. The frozen run contains 4,608 development experiments, 9,216 calibration experiments, and 9,216 evaluation experiments.

10.12 Five disclosed model views

The five models ask different questions rather than hiding one preferred representation behind a common label. Observation ridge receives twelve natural-state features and fits 78 coefficients.

Causal ridge receives twelve intervention-response features and fits 78. Joint ridge receives twenty-nine observed, causal, and carrier features and fits 180. The **animal-moment ridge** receives forty-eight means and dispersions and fits 294; it is not called a hierarchical model because it does not fit animal-level random effects. The sequence model passes each animal’s time series through one frozen 32-state reservoir, averages final and temporal-mean states into sixty-four features, and fits a 390-coefficient ridge readout. These are transparent synthetic baselines, not faithful implementations of biological astrocytes, canonical reservoirs, or the strongest available machine-learning systems.

For model (m), the standardized ridge readout remains

$$\widehat{W}_m = (X_m^\top X_m + \lambda P_m)^{-1} X_m^\top Y. \quad (26)$$

Different coefficient counts are reported rather than disguised as capacity matching. The relevant fairness condition in this gate is exact disclosure and common data exposure. A later full-model comparison would need architecture-native tuning, equalized resource contracts, and independent implementation.

10.13 Seed-block conformal role sets

Forced singleton decisions would repeat the paper’s original error by making uncertainty look like content. The factorial envelope therefore returns a set of roles when the evidence does not discriminate one role. For each model and factorial cell, calibration seed (s) contributes the maximum nonconformity across its six role rows,

$$U_{mcs} = \max_{r \in \mathcal{R}} [1 - p_m(r \mid x_{mcrs})]. \quad (27)$$

With (M=12) independent calibration seed blocks and (alpha=0.10), the cell-specific threshold is the finite-sample order statistic

$$q_{mc} = U_{mc(\min\{M, \lceil (M+1)(1-\alpha) \rceil\})}. \quad (28)$$

The reported role set is

$$\Gamma_{mc}(x) = \{r \in \mathcal{R} : 1 - p_m(r \mid x) \leq q_{mc}\}. \quad (29)$$

This block calibration targets simultaneous inclusion of all six correct roles within a future model-cell-seed block. It deliberately produces wider sets than row-wise calibration. Coverage without precision is therefore not treated as identification: singleton accuracy, simultaneous coverage, role-set size, wrong singletons, and false-content singletons are all retained.

10.14 Factor effects and multiplicity

For each model and evaluation seed, the main effect of factor (j) is the mean singleton-accuracy difference between its high and low levels after averaging over the other six factors,

$$\Delta_{mjs} = \overline{A}_{m,s}(F_j = H) - \overline{A}_{m,s}(F_j = L). \quad (30)$$

Two-factor interaction (j,k) is the paired difference in differences,

$$I_{mjs} = [\bar{A}(H, H) - \bar{A}(L, H)] - [\bar{A}(H, L) - \bar{A}(L, L)]. \quad (31)$$

Intervals are 95% (t) intervals over twelve seed-block estimates. Exact two-sided sign tests are Holm-adjusted across a frozen family of 35 model-by-main-effect tests and, separately, 105 model-by-interaction tests. An adjusted non-rejection is retained as non-rejection; it is not rewritten as absence of interaction.

10.15 Factorial results: success regions, failure regions, and reversals

No model is a universal role identifier. Mean singleton accuracy is 0.491 for observation ridge, 0.358 for causal ridge, 0.634 for joint ridge, 0.633 for animal-moment ridge, and 0.274 for the frozen-reservoir sequence model. The corresponding seed-block 95% intervals are 0.488-0.494, 0.356-0.360, 0.631-0.638, 0.628-0.637, and 0.271-0.277. These narrow intervals describe repeated synthetic seed blocks, not biological populations.

The averages conceal broad failure. Observation, causal, joint, animal-moment, and sequence models fall to one-sixth singleton accuracy or below in 64, 80, 40, 41, and 83 of 128 cells, respectively. Across all 640 model-by-cell combinations, 308 are at or below that chance reference, 232 return a mean role-set size of at least four, 38 contain at least one wrong singleton, and 11 contain at least one false-content singleton. The raw held-out predictions contain 51 wrong singletons and 14 false-content singletons. Every pair of model families reverses rank in at least one cell. Joint ridge exceeds animal-moment ridge in 16 cells, loses in 22, and ties in 90; their overall mean accuracies differ by only 0.0016. Even the weak sequence model exceeds each of the two strongest models in three registered cells.

Mean simultaneous seed-block coverage is 0.958 for observation, 0.947 for causal, 0.960 for joint, 0.965 for animal-moment, and 0.930 for sequence. Mean role-set sizes are 3.08, 3.40, 2.18, 2.18, and 3.46. Coverage therefore cannot be read as precise identification. Moreover, 18, 20, 15, 13, and 27 cells, respectively, fall below 0.90 simultaneous coverage. One technical correction is preserved openly: the frozen contract named simultaneous seed-block coverage, while the generator's first summary file reported marginal row coverage. A separate deterministic analysis computed the frozen simultaneous endpoint directly from the unchanged held-out covered/not-covered rows. No generator, model, threshold, prediction, family, or outcome was changed.

Twenty-three of 35 main effects survive the frozen Holm family, and eight of those are adverse. The largest favorable effects are observational separability for the observation model (+0.981), causal separability for the causal model (+0.717), and observational and causal separability for the joint model (+0.642 and +0.350). Intervention spread reduces causal-model accuracy by 0.207 and joint accuracy by 0.090. High measurement noise reduces causal, joint, animal-moment, and sequence accuracy by 0.283, 0.213, 0.216, and 0.418. The sequence model responds positively to intervention spread by 0.040 inside this generator, a model-specific reversal that rules out treating spread as uniformly harmful. Twelve main effects do not survive correction.

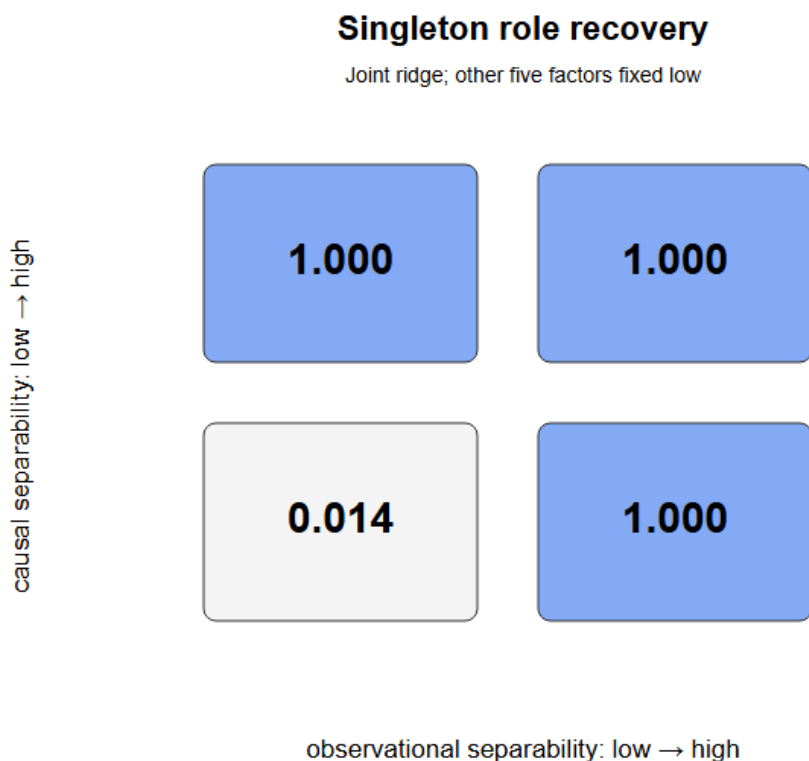
None of 105 two-factor interactions survives its separate Holm family at 0.05. The largest point estimates are not small: causal separability by noise is -0.567, observational by causal separability is -0.547 for joint ridge and -0.533 for animal-moment ridge, and causal separability by intervention spread is -0.413 for causal ridge. Their smallest adjusted value is 0.0513 because twelve paired seed

blocks provide a coarse exact-sign resolution under a 105-test family. The result is therefore **not** that interactions are absent. It is that this gate does not confirm any after its frozen multiplicity correction.

10.16 Carrier decomposition, figure, and exact replay

The factorial carrier construction succeeds at separating address overlap from task projection. Mean address Jaccard takes the designed extremes 0 and 1, while task-direction cosine takes approximately 0 and 0.88. Fixed-decoder accuracy ranges from 0.445 to 1.000; thirty-eight cells are at or below 0.50. Unlabeled relational alignment ranges from 0.997 to 1.000 and supplies as much as 0.555 gain, yet sixteen cells have no positive gain. That strong relational performance is a property of this transparent generator and its orthogonal alignment, not evidence that a brain uses the same operation.

Figure 4 shows one four-cell slice of the 128-cell design. When the other five factors are fixed low, joint-ridge singleton recovery is 0.014 with both observational and causal separability low, reaches 1.000 when either is high, and remains 1.000 when both are high. The slice illustrates the basic identifiability problem; it does not summarize the full seven-factor envelope.



The primary run and full replay reproduce all eight generated result files and the SVG byte-for-byte. The deterministic analysis and its replay reproduce five derived tables and their receipt byte-for-byte. Fifteen pre-outcome contract tests and eighteen post-run result-integrity tests pass. The adverse ledger retains 807 cell failures, null or adverse effects, non-rejected interactions, diffuse role

sets, wrong singletons, false-content singletons, and undercoverage findings. Fifty-four additional carrier-failure rows are retained. The final biological test remains sealed.

The factorial envelope also repairs the uncertainty unit used in the earlier demonstrations. The first and independent benchmarks reported Wilson intervals over role rows, even though the six roles generated under one experiment seed share nuisance state. Their point estimates remain unchanged, but the governing intervals are now 95% t intervals across 24 first-benchmark or 20 independent-benchmark seed-block accuracies. First-benchmark nominal intervals become 0.069-0.181 for observation, 0.882-0.965 for cross-perturbation, and 0.897-0.978 for the latent-surrogate model. Independent-benchmark nominal intervals become 0.946-1.000 for observation, 0.887-0.980 for causal endpoints, 0.252-0.448 for temporal dynamics, and 0.073-0.193 for the shuffled control. All 49 model-by-condition cluster summaries are retained and exactly replayed. This correction widens or shifts uncertainty appropriately without changing a prediction, model, rank, or point estimate.

11. Generator-family transport across continuous regimes

11.1 The ordinary problem: a model can recognize the laboratory instead of the role

The factorial envelope showed that role recovery changes across a seven-factor grid. That result still leaves a harder problem. All 128 cells belonged to one generator family with two declared levels per factor. A model could therefore learn the generator’s visual accent rather than the operation called content, index, gate, eligibility, stabilizer, or global state. Smooth interpolation inside the same family would not reveal that failure.

The biological analogue is immediate. One laboratory may tag a different astrocyte population, use a different promoter, sample at a different timescale, measure a different behavioral endpoint, or operate in a different circuit. If a classifier has learned those local regularities, high within-study accuracy can disappear when the preparation changes. The scientific problem is therefore not only whether roles are identifiable under favorable measurements. It is whether a role relation survives a change in how the observations are generated.

This concern strengthens rather than weakens the multicellular account. Recent primary studies further separate the operations that a memory experiment can measure. Astrocyte-specific deletion of ankyrin-2 in adult mice impaired remote while sparing recent memory, disrupted late-phase potentiation maintenance, reduced learning-dependent astrocyte contact with engram neurons, and altered astrocyte morphology without changing several basal electrophysiological measures [20]. Temporally precise tagging of dorsal-CA1 neurons identified nonoverlapping ensembles recruited during distinct periods of learning, with only some ensembles contributing to the core fear engram [21]. In macaques, older content-biased assemblies showed less drift, greater connectivity, stronger sleep reactivation, and persistent task-to-sleep coupling [22]. These findings make role, timescale, carrier, and measurement regime more important, not less.

Only after that problem is visible is the phrase **generator-family transport** useful. It means that an estimator fitted on two declared synthetic mechanisms is evaluated on a third mechanism whose rows were never used for fitting. It is not a claim that any generator faithfully reproduces an astrocyte or that transport among synthetic mechanisms establishes transport to biology.

11.2 Frozen continuous design

The generator-family transport study replaces the binary grid with forty-eight deterministic seven-dimensional Halton points. For point (i) and factor (j), the unit coordinate is

$$h_{ij} = \sum_{k=0}^{\infty} \frac{a_k(i + 23)}{p_j^{k+1}}, \quad (32)$$

where (a_k) are the base- (p_j) digits and $(p_j \in \{2, 3, 5, 7, 11, 13, 17\})$. Each coordinate is mapped before outcome access into a declared physical-analogy range: observational and causal separability span 0.05-1.00; address overlap spans 0-1; fixed-decoder projection spans -0.95-0.95; latent-neuronal loading spans 0-0.90; intervention spread spans 0-0.75; and measurement noise spans 0.06-0.52.

Three independently coded generators receive those coordinates:

1. an additive Gaussian family;
2. a nonlinear threshold family with sigmoid factor gains and coupled hyperbolic-tangent transforms; and
3. a rotated heteroskedastic family with independent role bases, role-varying gains, and coordinate-dependent noise.

For held-out family (g), the fitting and evaluation sets are

$$\mathcal{D}_{-g}^{train} = \bigcup_{g' \neq g} \mathcal{D}_{g'}^{dev}, \quad \mathcal{D}_g^{eval} = \mathcal{D}_g^{eval}, \quad \mathcal{D}_{-g}^{train} \cap \mathcal{D}_g^{eval} = \emptyset. \quad (33)$$

Development seeds 7000-7007, calibration seeds 7100-7111, and evaluation seeds 8000-8023 are disjoint. The frozen study contains 6,912 development rows, 10,368 calibration rows, 20,736 evaluation rows, and 82,944 held-out predictions. The runner does not import any generator from the preceding benchmarks or factorial envelope.

11.3 Four disclosed estimators rather than one preferred machine

Every estimator receives the same twenty-nine raw variables, but their capacities are not declared equal. Linear ridge uses twenty-nine standardized features and fits 180 coefficients. Quadratic ridge adds twenty-nine squares and 406 pair products, fitting 2,790 coefficients. Two frozen nonlinear maps retain the raw variables and add ninety-six cosine or tanh features, each fitting 756 coefficients. For model (m),

$$\phi_m(x) \in \mathbb{R}^{d_m}, \quad d_m \in \{29, 464, 125, 125\}, \quad (34)$$

followed by the same ridge readout used in the earlier studies. The comparison is common-exposure and capacity-reported, not capacity-matched. Across the three leave-one-family-out folds, 13,446 fitted coefficients and 1,102,464 raw feature values are explicitly accounted for. Runtime is recorded as execution metadata because wall-clock duration is hardware- and load-dependent; it is not inserted into the deterministic scientific digest.

11.4 Source calibration and target calibration answer different questions

Marginal conformal coverage can conceal failures in a particular class, group, or shifted distribution [23-25]. The generator-family transport study therefore carries two thresholds without refitting the estimator. Source calibration pools twenty-four complete source-family-by-seed blocks. Target calibration uses twelve new calibration blocks from the held-out generator. For calibration domain (d),

$$U_{mgs}^{(d)} = \max_{i,r} \left[1 - p_m(r \mid x_{g,s,i,r}^{(d)}) \right], \quad (35)$$

and the finite-sample threshold is

$$q_{mg}^{(d)} = U_{mg(\min\{M_d, \lceil (M_d+1)(1-\alpha) \rceil\})}^{(d)}. \quad (36)$$

The source threshold asks whether uncertainty transports without seeing the target family. The target threshold asks how wide a set becomes after a small, disjoint calibration sample from the target family is allowed. Neither threshold changes the fitted classifier. Coverage is reported simultaneously over all forty-eight points and six roles within each evaluation seed block. Wilson intervals over twenty-four block-level successes govern the bounded coverage readback; role and point rows are not counted as independent replications.

11.5 Continuous effect model and frozen multiplicity

Within each evaluation seed, role accuracy is averaged at every Halton point and fitted to seven standardized factors plus their twenty-one pair products:

$$A_{mgsi} = \beta_{0,mgs} + \sum_{j=1}^7 \beta_{j,mgs} z_{ij} + \sum_{j < k} \beta_{jk,mgs} z_{ij} z_{ik} + \epsilon_{mgsi}. \quad (37)$$

Each coefficient therefore has twenty-four independent seed-block estimates. Exact two-sided sign tests are Holm-adjusted across the frozen family of eighty-four main effects and, separately, 252 pair interactions. This greater resolution corrects the factorial envelope’s coarse twelve-seed interaction test. Rejection remains conditional on the declared synthetic family; non-rejection remains unresolved rather than becoming an absence claim.

11.6 Central adverse result: transport can reverse apparent identifiability

The additive and nonlinear families provide the favorable side of the study. Held-out additive accuracy ranges from 0.824 to 0.933 across the four estimators. Held-out nonlinear-threshold accuracy ranges from 0.814 to 0.887. Quadratic ridge is strongest in both, reaching 0.933 (95% seed-block interval 0.927-0.939) and 0.887 (0.882-0.893).

The rotated heteroskedastic family reverses that picture. Linear, quadratic, cosine-feature, and tanh-feature accuracy falls to 0.024, 0.048, 0.026, and 0.036. Every value is far below the one-sixth six-role singleton reference. This is not evidence that the biological roles are unlearnable. It is direct

evidence that the apparent role decoder learned basis-dependent regularities that did not survive the independent rotation and noise law.

Define the transport loss relative to the mean of the two source-like held-out families as

$$L_{m,g}^{transport} = \frac{1}{2} \sum_{g' \neq g} A_{m,g'} - A_{m,g}. \quad (38)$$

For the rotated family this loss is large for every estimator, despite their different feature maps and fitted capacities. Quadratic ridge remains numerically strongest at 0.048, showing that a model can win every registered family while still fail absolutely. One model-pair ranking reverses: linear slightly exceeds the cosine-feature model in the additive and threshold families, while the cosine-feature model slightly exceeds linear in the rotated family. Seventy-five factor or interaction terms also reverse coefficient direction between generators.

The strongest objection is that the rotated family deliberately changes the coordinate basis through which a role appears. That objection is correct but does not dissolve the result. The generator-family transport study tests adapter-free transport: no labeled target row and no target-derived alignment may alter the estimator. The result does **not** show that harmonization, unlabeled geometric alignment, domain adaptation, or a small labeled target sample could never recover the roles. It shows that the frozen decoder did not identify a coordinate-free role relation. A later gate must compare no adaptation, unlabeled alignment, labeled calibration, and full refitting as separate interventions rather than silently granting the target basis to the preferred model.

11.7 Coverage recovered by near-total ambiguity is not identification

Source calibration fails simultaneous 90% coverage in nine of twelve held-out-family/model rows. The most severe cases occur under the rotated generator, where source-calibrated simultaneous coverage is zero for all four estimators. Target calibration raises rotated-family coverage to 0.958, 0.750, 0.958, and 0.917, but the corresponding role sets average 5.9988-5.9996 of the six possible labels. The classifier is therefore covered because it nearly always says "any role," not because it recovers the hidden role.

The same precision problem appears elsewhere. Every one of the twelve target-calibrated rows has a mean set size of at least four. Three still fall below 90% simultaneous coverage after target calibration. The nonlinear quadratic model rises from 0.083 source-calibrated to 0.708 target-calibrated coverage while its mean set expands from 3.83 to 5.29 roles. Four wrong source-calibrated singletons are retained; no false-content singleton appears in this particular study.

A useful calibration score must therefore retain both coverage and informativeness. One transparent descriptive quantity is

$$Q_{mg} = C_{mg} \left(1 - \frac{|\overline{\Gamma_{mg}}| - 1}{|\mathcal{R}| - 1} \right), \quad (39)$$

where (C) is simultaneous block coverage and the second factor falls from one for singleton sets to zero for six-role sets. This score is descriptive and was not a frozen primary endpoint. It makes the logical point visible: coverage purchased by total ambiguity supplies no role identification.

11.8 Continuous effects, interactions, and generator dependence

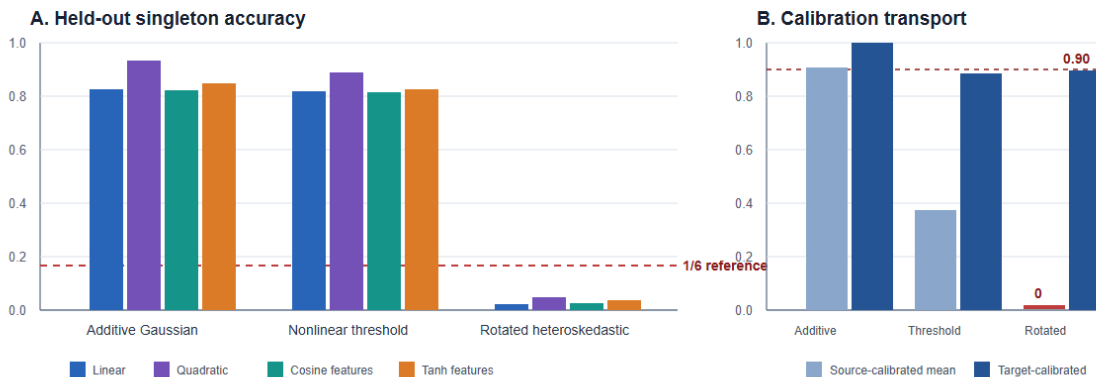
Thirty-three of eighty-four main effects and forty of 252 interactions survive their separate frozen Holm families. The strongest main effects are scientifically coherent within the declared mechanisms: noise is adverse in the additive and nonlinear families, while observational and causal separability are often favorable. For the nonlinear cosine-feature estimator, measurement-noise coefficient is -0.361; for the nonlinear linear estimator it is -0.358. Observational separability reaches +0.237 for the nonlinear tanh estimator. The minimum attainable all-same-sign (p) value with twenty-four seeds is $(2/2^{24})=1.19\times10^{-7}$.

The interaction findings also matter because the factorial envelope confirmed none. In the nonlinear family, observational-by-causal separability is negative across all four estimators (-0.109 to -0.148), and latent-neuronal-loading by noise is likewise negative (-0.120 to -0.123). These do not mean that observational and causal evidence oppose one another in brains. They mean that the nonlinear generator has saturation and confounding regimes in which adding one separability source supplies less incremental recovery when another is already strong. Seventy-five cross-generator direction reversals prevent these coefficients from being promoted into universal mechanisms.

The required adverse, null, and uncertain ledger contains 334 rows. It retains four below-reference transport summaries, nine source-undercoverage summaries, three target-undercoverage summaries, twelve diffuse target-set summaries, four wrong source singletons, every adverse rejected effect, and every multiplicity non-rejection. The primary and replay runs reproduce all eight scientific output files byte-for-byte. The deterministic descriptive analysis also replays exactly. Fifteen pre-outcome contract tests and fifteen post-outcome result tests pass. The biological final test remains sealed.

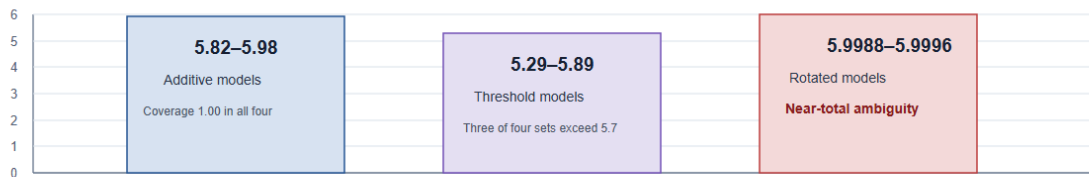
A strong in-family decoder can collapse under a new measurement law

Leave-one-generator-family-out role identification; 24 evaluation seed blocks per model and held-out family



C. Target calibration restores broad coverage mostly by returning almost every role

Mean target-calibrated role-set size; six roles are possible



Synthetic result only. It tests estimator identifiability and calibration under declared generator shifts; it does not identify a biological astrocyte role.

11.9 Consequences for the biological experiment

The 2026 biological literature points toward a design that tests transport rather than assuming it. Ank2 loss separates recent from remote memory and early from maintained LTP while also changing astrocyte morphology [20]. Temporally precise neuronal tagging separates ensembles recruited during different learning events [21]. Primate recordings show that memory age, drift, connectivity, sleep reactivation, and assembly integration change together [22]. A strong multicellular-engram test should therefore freeze the following before outcome access:

1. a role definition tied to content, access, stability, precision, eligibility, or general state;
2. at least one transfer across animal, day, circuit, tool, or laboratory;
3. complete animal-level blocks as the uncertainty unit;
4. source-only calibration and a separately labeled target-calibration analysis;
5. role-set size, wrong singleton, and false-content risk alongside coverage;
6. cell-type and intervention-footprint validation;
7. recent-versus-remote, encoding-versus-recall, and fast-versus-slow timescale contrasts;
8. an unresolved disposition when transport or precision fails.

Those primary studies also impose boundaries. The Ank2 behavioral experiments reported male mice and a particular set of genetic, viral, electrophysiological, and behavioral tools [20]; they support a persistence mechanism, not a universal astrocyte-content code. The temporally decomposed CA1 study is neuronal [21], and the macaque assembly study does not identify an astrocyte carrier [22]. They strengthen the need to separate roles, timescales, and carriers. They do not retroactively validate SAN terminology or transfer their results to every memory system.

The generator-family transport study failure supports the paper’s original role-separation argument in a more demanding form. Content language cannot be earned by cue correlation, by perturbation success, by within-generator accuracy, or by high coverage alone. The role must remain identifiable across a declared change in measurement conditions, and the uncertainty set must remain informative. If it does not, the correct result is not "astrocytes contain the memory" or "astrocytes do not contain the memory." The correct result is that this experiment has not separated the candidate roles.

11.10 What the transport study establishes

The generator-family transport study advances the paper from a binary within-family envelope to continuous, leave-one-generator-family-out transport. Its most important result is adverse: three different model views can look strong in familiar synthetic regimes and collapse below chance after an independently rotated measurement law. Target calibration can restore nominal coverage while returning almost every possible role. The result therefore identifies a necessary standard for the biological program: transport and set precision must accompany accuracy and coverage.

This study still does not identify a biological astrocyte role. It does not establish that a real experiment will undergo the synthetic rotation, and it does not validate the virtual-engram or receiver-relative PWD extensions. The next necessary step is a prospective multilevel biological design and simulator that represents animals, sessions, laboratories, cell-type specificity, intervention footprints, memory age, and recent-versus-remote endpoints; it must preserve the present adverse transport result and keep the biological test sealed.

12. Multilevel biological design across animals, sessions, and laboratories

12.1 The ordinary problem: the same intervention is not the same experiment everywhere

A memory experiment does not produce one independent row at a time. The same animal contributes several sessions. Recent and remote recall occur after different intervals. Encoding and retrieval manipulations ask different causal questions. A viral promoter, capsid, titer, ligand, reporter, microscope, behavioral assay, and laboratory workflow each change what is labeled, changed, or measured. If those conditions are collapsed into one flat table, an estimator can confuse a laboratory signature with a memory role and can count repeated observations from one animal as if they were independent replications.

The biological literature makes these distinctions concrete. Neocortical and hippocampal engram participation changes across systems consolidation, so recent and remote recall cannot be treated as interchangeable measurements [26]. Astrocyte-directed viral expression can include off-target neuronal expression, and promoter sequence alone may not guarantee cell-type restriction [27]. A later microRNA de-targeting system substantially improved astrocytic specificity across several constructs and serotypes, yet specificity still depended on titer, delivery, serotype, region, and reporter configuration [28]. Chemogenetic intervention also has a footprint beyond the engineered receptor: CNO can act after conversion to clozapine, and CNO administration has produced behavioral effects in control animals under some conditions [29,30]. These studies do not make astrocyte experiments uninterpretable. They show why cell identity and intervention spread must be measured rather than assumed.

Only after this problem is stated is the term **multilevel biological design** useful. It means that the experimental unit, repeated session, laboratory, tool, promoter, memory age, intervention timing, specificity, and footprint are represented separately before an outcome is interpreted. The multilevel biological-design study is still synthetic. Its purpose is to test whether the paper’s role-identification rules remain coherent under that structure before any sealed biological endpoint is opened.

12.2 Frozen animal-session-laboratory structure

The pre-outcome contract defines four synthetic laboratories, thirty-six animals per laboratory, and three disjoint animal sets: twelve development animals, eight calibration animals, and sixteen evaluation animals. Each animal contributes four sessions formed by recent versus remote memory and encoding versus recall intervention timing. Every session contains one row for each of the six candidate roles. The complete simulator therefore contains 3,456 feature rows. Across four held-out-laboratory folds and four declared access conditions, it produces 6,144 evaluation predictions.

For role (r), animal (a), session (s), tool (t), promoter (p), and laboratory (l), the synthetic observation has the form

$$x_{lastrp} = D_l T_l [g_{larp} \mu_r + u_a + v_{as} + m_s q_r + f_{latp} h_r] + b_p + \epsilon_{lastrp}, \quad (40)$$

where μ_r is a role prototype, (u_a) is a persistent animal state, ($v_{\{as\}}$) is a session state, (m_s) encodes recent/remote and encoding/recall structure, (f) is intervention footprint, (T_l) is

the laboratory measurement basis, (D_l) is a laboratory-specific scale, and (b_p) is a promoter-associated observation shift. Cell-type specificity changes the recoverable role gain, while intervention footprint adds a global component and leakage toward a neighboring role:

$$g_{larp} = 1.05 [0.58 + 0.52 c_{lap}] [1 - 0.34 f_{latp}] \rho_{rs}, \quad c_{lap} \in [0.50, 0.99], \quad f_{latp} \in [0.02, 0.75]. \quad (41)$$

The role- and session-specific multiplier ρ_{rs} makes stabilization more prominent in the remote condition, index modestly more prominent remotely, gate more prominent at recall, and eligibility more prominent at encoding. These coefficients define the simulator; they are not fitted biological effect sizes. The tool and promoter labels are neutral design factors. They do not report the measured performance of a named commercial vector or product.

Laboratories A and B are small perturbations of a shared observation basis. Laboratory C has a larger perturbation and a bounded nonlinear compression. Laboratory D has an independently constructed observation basis, greater noise, and lower mean specificity. The design therefore includes familiar and unfamiliar measurement laws without asserting that these four cases estimate how often real laboratories will agree.

The uncertainty unit is the complete evaluation animal, not one role row, one session, or one probability. Each animal block contains twenty-four rows: two memory ages, two intervention times, and six roles. All intervals and paired contrasts use the sixteen complete evaluation-animal blocks in a held-out laboratory.

12.3 Four information-access conditions

The generator-family transport study established that the word *adaptation* could conceal fundamentally different uses of the target domain. The multilevel biological-design study therefore freezes four non-interchangeable conditions. Their allowed target information is

$$\mathcal{I}_k = \begin{cases} \emptyset, & k = \text{no adaptation,} \\ \{X_{target}^{dev}\}, & k = \text{unlabeled alignment,} \\ \{Y_{target}^{cal}\}, & k = \text{labeled calibration,} \\ \{X_{target}^{dev}, Y_{target}^{dev}, Y_{target}^{cal}\}, & k = \text{full refit.} \end{cases} \quad (42)$$

No adaptation fits a source model and a source animal-block threshold, then applies both unchanged. Unlabeled alignment may inspect 288 target-development feature rows but no target label. It aligns only the eighteen-dimensional observation block by source and target means and covariances:

$$\tilde{X}_t = (X_t - \bar{X}_t) (C_t + \lambda I)^{-1/2} (C_s + \lambda I)^{1/2} + \bar{X}_s. \quad (43)$$

This is one disclosed covariance adapter. Its result cannot represent every label-free alignment method. Labeled calibration leaves the source classifier unchanged and uses 192 labeled rows from eight target calibration animals only to choose the prediction-set threshold. Full refit uses 288 labeled target-development rows to fit a new model and 192 labeled target-calibration rows to set its threshold. Full refit is therefore supervised target recovery. Calling it transport would erase the very distinction the experiment is designed to test.

All four conditions use the same twenty-eight-variable multivariate ridge form,

$$\hat{B}_k = \arg \min_B \|Y_k - [\mathbf{1}, Z_k]B\|_F^2 + \lambda \|B_{-0}\|_F^2, \quad (44)$$

with six outputs and 174 fitted coefficients per execution. The sixteen fold-condition executions materialize 2,784 coefficient entries. This is not a capacity comparison: the source model is deliberately repeated across some conditions so that information access, not estimator family, is the changing variable.

12.4 Animal-block calibration and evaluation

For each calibration animal (a), nonconformity is maximized over every session and role from that animal:

$$U_a = \max_{s,r} [1 - \hat{p}(r \mid x_{asr})], \quad q = U_{(\min\{M, \lceil (M+1)(1-\alpha) \rceil\})}. \quad (45)$$

Source calibration uses twenty-four animals from three laboratories. Target calibration uses eight animals from the held-out laboratory. With eight blocks and $\alpha = 0.10$, the finite target threshold is the maximum calibration-block score. That coarse resolution is disclosed rather than hidden.

Coverage is earned only if the prediction set contains every true role in every row from an evaluation animal:

$$C_{alk} = \mathbf{1}[\forall(s, r), r \in \Gamma_{lk}(x_{asr})]. \quad (46)$$

The primary readback reports singleton accuracy, mean animal-level simultaneous coverage, mean set size, wrong singletons, and false-content singletons. Recent/remote, encoding/recall, tool, promoter, and role summaries remain conditional descriptions. Paired adaptation contrasts are computed within animal,

$$\Delta_{alkm} = M_{alkm} - M_{al, \text{none}, m}, \quad (47)$$

for accuracy, simultaneous coverage, and mean set size. Two-sided exact sign tests are Holm-adjusted across the frozen family of thirty-six fold-condition-metric contrasts. Non-rejection remains unresolved; it is not converted into equivalence or absence.

12.5 Main result: access can repair point accuracy without repairing calibrated transport

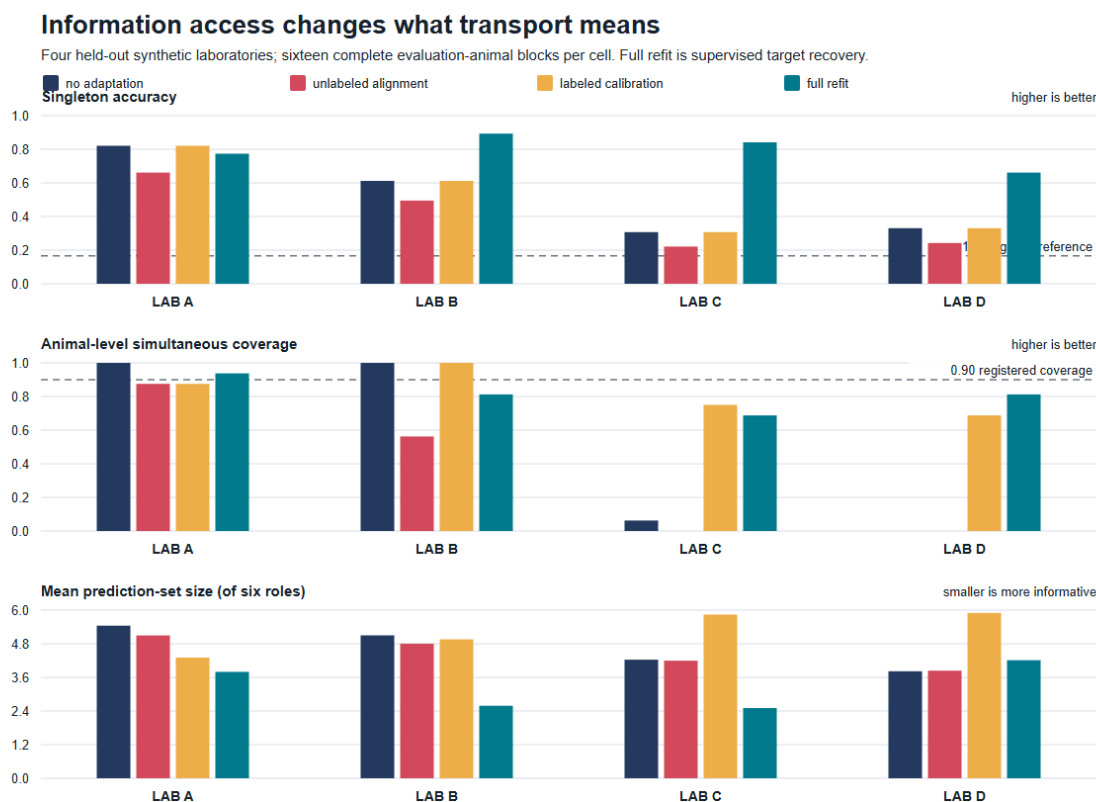
Adapter-free singleton accuracy is 0.820 in laboratory A, 0.612 in B, 0.307 in C, and 0.331 in D. These values are all above the one-sixth singleton reference, but they do not establish useful transport. Source-calibrated simultaneous coverage is 1.000 in A and B, 0.063 in C, and zero in D. The apparently favorable coverage in A and B is purchased with mean prediction sets of 5.45 and 5.10 roles. In other words, the source model often covers by naming nearly every possible role.

Unlabeled covariance alignment does not repair the problem in this simulator. Accuracy falls to 0.661, 0.495, 0.221, and 0.242 across laboratories A-D. The paired harm is Holm-significant in A,

B, and C; the D contrast does not survive the full multiplicity family. Coverage is 0.875, 0.563, zero, and zero. Mean sets remain 5.10, 4.80, 4.20, and 3.84 roles. This is an adverse result for the registered covariance adapter, not a claim that every unlabeled adapter must fail.

Target-label calibration cannot change a point prediction because the classifier is frozen. Its accuracy is therefore exactly the no-adaptation accuracy in all four folds, an equality verified row by row. It changes uncertainty instead. Coverage becomes 0.875, 1.000, 0.750, and 0.688, while mean set size becomes 4.31, 4.96, 5.84, and 3.84. In C and D, the large coverage gain is mainly purchased by returning almost every role. Labeled calibration is useful for revealing uncertainty, but it cannot manufacture a target-valid decision boundary.

Full refitting increases accuracy strongly in laboratories B-D: 0.893 in B, 0.841 in C, and 0.661 in D, with Holm-significant paired gains of +0.281, +0.534, and +0.331. Laboratory A moves from 0.820 to 0.773, a non-significant decrease of 0.047. Supervised target recovery therefore helps where the source boundary is poorly oriented, but it is not uniformly beneficial in a small finite development set. More importantly, full-refit animal-level simultaneous coverage is 0.938, 0.813, 0.688, and 0.813. Only laboratory A reaches the registered 0.90 endpoint. Strong point accuracy and compact average sets do not guarantee that every condition within an animal is covered.



12.6 Recent and remote memory remain a designed contrast, not a retrospective explanation

The simulator contains an explicit recent/remote mechanism, but the measured differences remain conditional on laboratory and access. Remote-minus-recent accuracy is positive in many cells, negative in the no-adaptation and aligned laboratory-C cells, and near zero in five registered

condition-fold summaries. Full-refit accuracy changes from recent to remote by +0.057 in A, +0.036 in B, +0.016 in C, and +0.042 in D. These values demonstrate that the pipeline can retain the contrast without forcing a universal direction.

The appropriate biological experiment should be stricter. Recent and remote endpoints should be declared before data access; animals should remain the uncertainty unit; neuronal activity, astrocytic state, and behavioral expression should be measured together; and a persistence claim should require an interaction between memory age, intervention timing, and the proposed cellular role. A main effect of reduced freezing, slower movement, altered arousal, or general performance cannot by itself identify content, indexing, gating, eligibility, or stabilization.

The same discipline applies to the 2026 Ank2 result that motivated this design. It provides a strong persistence and morphology mechanism in a bounded mouse preparation [20]. It does not mean every astrocyte population stores remote memory content. The 2017 systems-consolidation study likewise supports changing recent/remote neuronal circuits [26]; it does not transfer its carrier identity to astrocytes. The sources jointly justify the contrast while preserving their different cellular and experimental scopes.

12.7 A discovered allocation limitation and its prospective repair

The multilevel biological-design study simulator successfully represents tool and promoter labels, but the deterministic allocation aliases them within each held-out evaluation laboratory. Each laboratory contains only three of the nine possible tool-by-promoter combinations. Tool-conditional and promoter-conditional accuracy ranges are therefore numerically confounded and cannot be interpreted as separate effects. Cell-type specificity and intervention footprint are also partly generated from those assignments, so their observed associations are design variables rather than causal estimates.

The result is not retrospectively reinterpreted. The frozen runner remains unchanged, and the limitation is preserved in a seven-row design ledger. The prospective repair fully crosses tool and promoter within laboratory and split, independently varies specificity and footprint, and retains enough animals per crossed cell for animal-level uncertainty. It should also compare more than one unlabeled adapter. The multilevel biological-design study therefore advances the hierarchy and access-control logic while refusing to claim that it has independently identified tool, promoter, specificity, or footprint effects.

12.8 Every adverse transport result remains governing

The multilevel biological-design study does not replace the generator-family transport study counterexample. The exact 334-row adverse/null/uncertain ledger remains hash-bound. It retains four below-reference rotated-family accuracy summaries, nine source-calibration undercoverage summaries, three target-calibration undercoverage summaries, twelve diffuse target-set summaries, four wrong source singletons, the bounded zero false-content count, 263 multiplicity non-rejections, forty-one rejected adverse effects, seventy-five coefficient-direction reversals, and one model-pair rank reversal. The relevant files are carried forward by exact SHA-256 rather than summarized into oblivion.

The multilevel biological-design study adds sixty-four registered adverse, null, ambiguity, and target-dependence rows: twelve undercoverage rows, eleven diffuse-set rows, eight target-label-dependent

rows, eighteen multiplicity non-rejections, five accuracy harms, four ambiguity increases, five near-null recent/remote contrasts, and one wrong singleton. No false-content singleton occurs in this finite gate. That zero is retained as a bounded count, not a zero-risk conclusion.

The primary and replay executions reproduce ten scientific files byte-for-byte. The derived analysis and its replay reproduce four files byte-for-byte. Twenty-one pre-outcome contract tests and twenty-nine post-outcome result tests pass. These checks prove execution, access-policy enforcement, adverse-result retention, and deterministic replay within the declared program. They do not prove the biological hypothesis.

12.9 Consequences for the decisive biological experiment

The simulator clarifies what must be earned before content language is appropriate. A biological study should require all of the following:

1. fully crossed or explicitly estimable tool, promoter, dose, and delivery conditions;
2. quantitative cell-type specificity and spatial intervention footprint for every animal;
3. disjoint development, calibration, and evaluation animals;
4. recent and remote recall with encoding and recall perturbations;
5. simultaneous neuronal, astrocytic, physiological, and behavioral measurements;
6. source-only transport, unlabeled adaptation, labeled calibration, and supervised refit reported as different questions;
7. animal-level simultaneous coverage and prediction-set precision alongside point accuracy;
8. neuronal bypass or rescue capable of separating an astrocyte-conditioned route from a hidden neuronal content trace;
9. matched ligand, vector, promoter, movement, arousal, and general-state controls;
10. an unresolved disposition when tool identity, carrier identity, or role remains aliased.

An informative result would be more specific than a behavioral change. For example, a stabilizer disposition would require a selective deficit in persistence or remote access, preserved content when retrieval is externally supported, a measured astrocytic state change, and a neuronal account that cannot explain the effect alone. A content disposition would require content-specific information in the manipulated cellular state, selective perturbation of that information, and a matched rescue that reinstates the same remembered relation without merely changing arousal or permissive gain.

The SAN receiver-relative extension remains one model beyond this role taxonomy. It would have to show that a tonic contextual state and a receiver-relative phase residual improve held-out prediction or rescue after simpler neuronal, global-state, access, and stabilization variables are included. The multilevel biological-design study contains no biological phase measurement and grants no support to that extension. Preserving the full hypothesis means preserving the test that could distinguish it, not declaring it from synthetic role recovery.

12.10 What the multilevel study establishes

The multilevel biological-design study advances the paper from generator-family transport to a prospective animal-session-laboratory architecture. Its central result is not that adaptation solves transport. Unlabeled covariance alignment harms accuracy in all four folds; labeled calibration often

recovers coverage by expanding uncertainty sets; and supervised refitting improves point accuracy in three difficult targets while missing the registered simultaneous-coverage endpoint in three of four laboratories. The study also exposes a tool-promoter allocation alias that prevents separate causal interpretation.

The result is therefore a successful methods analysis with an adverse design discovery, not a biological validation. The preceding studies remain preserved by exact hash. The final biological test remains `SEALED_NOT_OPENED`. The next necessary step is a preregistered, fully crossed tool-by-promoter multilevel simulator with independently varied specificity and footprint, more than one unlabeled adapter, and power or precision calculations tied to the eventual animal experiment. Only after that repair should the program freeze a biological protocol or open any final endpoint.

13. Fully crossed causal identifiability and power

13.1 The ordinary problem: naming two factors does not separate their effects

The multilevel biological-design study named tool and promoter, but a name in a table is not an estimand. Within each laboratory, only three of the nine possible tool-by-promoter combinations appeared. A higher score associated with one tool could therefore have been produced by its paired promoter, its paired specificity range, its paired footprint, or the unmeasured mixture of all four. The correct response was not to reinterpret the result. It was to preserve the alias and build a new prospective design in which the alternatives can actually be separated.

This is the experimental-design problem made classical by Fisher: replication does not repair a confounded allocation, while crossing and randomization create comparisons that can answer more than one-factor-at-a-time questions [31]. Modern causal language adds a second discipline. Positivity, exchangeability, consistency, no interference, and measurement validity are assumptions that connect an allocation to a causal interpretation; a full-rank matrix is necessary for the present estimands but cannot establish every assumption by itself [32]. Only after those requirements are visible is **fully crossed causal identifiability** a useful name for this design repair.

13.2 The prospective allocation repair

The fully crossed design study represents three tools and three promoters in every development, calibration, and evaluation split of every synthetic laboratory. If $(N_{l,s}\{t,p\})$ is the number of complete animal blocks assigned to tool (t) and promoter (p) in laboratory (l) and split (s) , the frozen positivity rule is

$$N_{l,s}(t, p) = n_s > 0 \quad \text{for every } (l, s, t, p), \quad n_s \in \{4, 3, 6\} \quad (48)$$

for development, calibration, and evaluation respectively. The design contains 36 development, 27 calibration, and 54 evaluation animals per laboratory: 468 complete synthetic animal blocks in total. Each animal contributes recent and remote memory sessions crossed with encoding and recall intervention timing, and each session contains all six candidate roles. The feature table therefore contains 11,232 rows.

Cell-type specificity and intervention footprint no longer inherit their values from promoter and tool. The same stratified specificity levels and the same stratified footprint levels occur inside every

tool-by-promoter cell, with a cyclic schedule keeping their observed correlation at or below 0.1112 in every laboratory-split block. For the eleven-column allocation matrix containing an intercept, eight cell indicators, specificity, and footprint,

$$\text{rank}(X_{l,s}) = 11 \quad \text{for all twelve laboratory-split blocks.} \quad (49)$$

Every block contains all nine cells at exactly the registered replication. This repairs the multilevel biological-design study’s structural alias inside the simulator. It does not prove that a real promoter, vector, ligand, reporter, region, or assay has been measured without error.

The frozen contract contained one arithmetic defect that is preserved rather than hidden. It stated 24,192 held-out predictions while correctly specifying 54 evaluation animals, four sessions, six roles, four laboratories, and seven conditions. The implemented and structurally correct count is

$$4 \times 54 \times 4 \times 6 \times 7 = 36,288. \quad (50)$$

Both executions retained all 36,288 rows. The original contract remains byte-identical, and an additive count erratum records the mistake, its correction, and the fact that no estimator, endpoint, threshold, allocation, or result changed.

13.3 Estimands before models

Let $(Y_i(t,p,c,f))$ denote a complete-animal role-recovery outcome under tool (t), promoter (p), specificity (c), and footprint (f). With tool 0 and promoter 0 as references, the two tool main effects are defined by averaging over all three promoter cells,

$$\tau_j^T = \frac{1}{3} \sum_{p=0}^2 \mathbb{E}[Y_i(j,p,C,F) - Y_i(0,p,C,F)], \quad j \in \{1,2\}, \quad (51)$$

and promoter main effects are defined symmetrically. The four interaction estimands are difference-in-differences,

$$\tau_{jp}^{TP} = \mathbb{E}[Y_i(j,p) - Y_i(j,0) - Y_i(0,p) + Y_i(0,0)], \quad j,p \in \{1,2\}. \quad (52)$$

These eight contrasts form the registered factorial family. Specificity and footprint receive a separate two-slope family conditional on the nine allocation cells,

$$\mathbb{E}[Y_i \mid T_i, P_i, C_i, F_i] = \alpha_{T_i P_i} + \beta_C(C_i - 0.76) + \beta_F(F_i - 0.36). \quad (53)$$

The synthetic allocation supplies positivity and declared exchangeability by construction. A biological interpretation would additionally require well-defined intervention versions, measured cell identity and spread, no between-animal interference, valid outcome measurement, complete-block retention, and an analysis that respects laboratory, animal, session, and time. The fully crossed design study identifies these estimands only inside its declared generator.

13.4 Two unlabeled adapters and a capacity-matched nonlinear comparator

The multilevel biological-design study tested one covariance adapter. The fully crossed design study retains it and adds a marginal empirical-quantile adapter. Both may inspect target-development features but no target label. They remain distinct from target-label calibration, which can change only the conformal threshold, and from supervised refitting, which uses target-development labels to change the decision boundary.

The linear and nonlinear readouts now fit equal capacity. After standardization ($z=(x-\bar{x})/s_x$), the nonlinear comparator applies one fixed pre-outcome orthogonal map (R) followed by a bounded nonlinearity,

$$\phi_{NL}(x) = \tanh(zR) \in \mathbb{R}^{28}, \quad \phi_L(x) = z \in \mathbb{R}^{28}. \quad (54)$$

Each basis then fits the same intercept-plus-28-feature, six-output ridge readout: 174 coefficients per execution. Equal fitted dimension does not make the models functionally identical or prove architecture optimality. It removes the simpler explanation that one readout won merely because it fitted more coefficients.

For the second label-free adapter, target observation (x_j) is mapped through its empirical target-development quantile and then through the source empirical inverse distribution,

$$\tilde{x}_j = \hat{F}_{S,j}^{-1}(\hat{F}_{T,j}(x_j)), \quad j = 1, \dots, 18. \quad (55)$$

This is a disclosed marginal alignment, not a general solution to domain adaptation. CORAL can use covariance geometry that the marginal map ignores; the marginal map can tolerate some coordinatewise distribution changes that CORAL does not. A failure of either is method-specific.

13.5 Executed results: the alias is repaired, but transport remains conditional

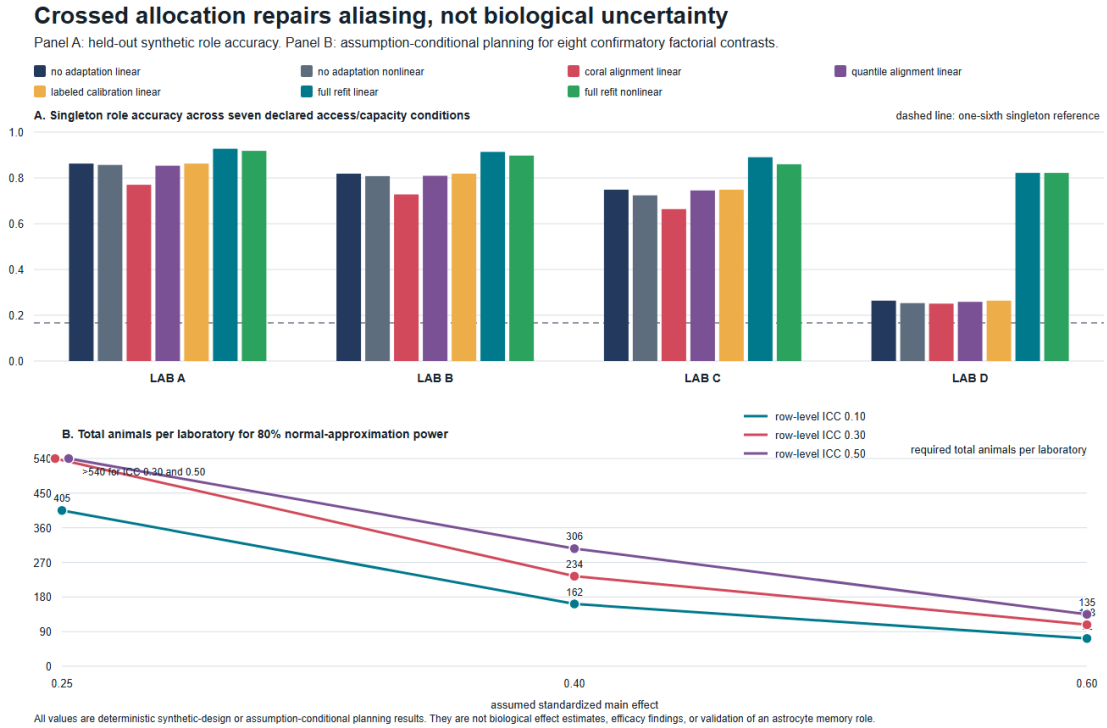
The source-only linear accuracy is 0.863, 0.819, 0.748, and 0.264 in laboratories A-D. CORAL alignment lowers those values to 0.770, 0.728, 0.664, and 0.251. The paired harms survive the frozen 84-test Holm family in A-C but not D. Empirical-quantile alignment yields 0.853, 0.809, 0.745, and 0.258; none of its accuracy differences from source-only linear survives that family. Two label-free methods therefore fail in different ways, and neither repairs the independently rotated laboratory D.

Supervised target refitting raises linear accuracy to 0.927, 0.914, 0.890, and 0.822; every gain over source-only linear survives multiplicity. That is evidence that the synthetic target contains recoverable role structure after labeled boundary repair. It is not label-free transport. The equal-capacity nonlinear refit reaches 0.918, 0.897, 0.860, and 0.822. It never improves significantly over the linear refit and is significantly worse in laboratory C by 0.0309 (Holm-adjusted ($p=0.00937$)). Equal capacity therefore does not turn nonlinearity into a default advantage.

Coverage must still be read with precision. In laboratory D, target-label calibration raises simultaneous coverage from zero to one while expanding the mean prediction set from 3.82 to 5.95 of six roles. That result is target-label dependent and nearly uninformative about which role is present. Full refitting raises laboratory-D coverage to 0.926 with mean sets of 3.40 for linear and 3.68 for nonlinear readouts. Across all 28 primary cells, nine remain below the registered 0.90 simultaneous-coverage endpoint and fifteen return mean sets of at least four roles.

The fully crossed allocation makes the eight tool/promoter contrasts estimable, but none of the 224 laboratory-condition-factorial readbacks survives its global Holm family. This is not proof of no factor effects. The simulator deliberately injects small tool, promoter, and interaction components, yet finite animal-level role-recovery accuracy does not resolve them under the registered multiplicity. The result is an adverse power and measurement finding: removing an alias is necessary but not sufficient for sensitive identification.

Specificity and footprint are also separately estimable. Thirty-six of 56 continuous-slope readbacks survive their separately frozen global Holm family, while twenty do not. Within the declared generator, higher specificity is often favorable and larger footprint often adverse. Those directions arise from the frozen simulator equations. They are not empirical effect estimates for a vector, promoter, drug, mouse, circuit, or human memory.



13.6 Power and precision belong to the design, not the conclusion

The fully crossed design study adds a prospective operating-characteristic map for the eventual eight-contrast tool/promoter family. If a row-level standardized outcome has intraclass correlation ρ across four sessions, the variance of one complete-animal mean is $(V_A = \rho + (1 - \rho)/4)$. With (n) animals in each of the nine tool-by-promoter cells, a main-effect difference averaged over three cells has

$$SE(\hat{\tau}) = \sqrt{\frac{2V_A}{3n}}, \quad z_* = \Phi^{-1}\left(1 - \frac{0.05}{2 \times 8}\right), \quad (56)$$

and the two-sided normal-approximation power under standardized effect δ is

$$\pi(\delta, \rho, n) = 1 - \Phi\left(z_* - \frac{\delta}{SE}\right) + \Phi\left(-z_* - \frac{\delta}{SE}\right). \quad (57)$$

The current evaluation design has six animals per cell. Under the registered assumptions, 80% power for a 0.60 standardized main effect requires eight animals per cell at $\rho = 0.10$, twelve at $\rho = 0.30$, and fifteen at $\rho = 0.50$. A 0.40 effect requires 18, 26, or 34 animals per cell. A 0.25 effect requires 45 animals per cell at $\rho = 0.10$ and more than sixty at $\rho = 0.30$ or 0.50 . These are planning consequences of assumed scales, not estimates of biological variance or effect size. A real protocol must replace every assumed input with pilot or external data before choosing a sample size.

Holm’s sequential procedure governs the 84 executed condition contrasts and the two synthetic estimand families [33]. The prospective power map uses the more conservative two-sided Bonferroni critical value for eight confirmatory factorial contrasts. Non-rejection remains unresolved. It cannot be promoted into equivalence, absence, or a claim that a tool or promoter is inert.

13.7 Adverse, null, and correction results remain visible

The fully crossed design study adverse/null/uncertain ledger contains 374 rows: 224 factorial multiplicity non-rejections, 45 planned-contrast non-rejections, twenty continuous-estimand non-rejections, twenty near-null recent/remote contrasts, fifteen accuracy harms, fifteen diffuse prediction-set results, nine simultaneous-undercoverage results, twelve target-label-dependent primary results, eight target-feature-dependent primary results, and six ambiguity increases. No wrong singleton or false-content singleton occurs in this finite study; both zeros remain bounded counts rather than safety claims.

The generator-family transport study 334-row adverse ledger and the multilevel biological-design study 64-row adverse ledger remain governing by exact hash. The multilevel biological-design study’s seven design limitations also remain preserved, including its four laboratory-specific alias rows. The fully crossed design study repairs that allocation prospectively; it does not rewrite the earlier result as if the defect never existed.

A separate implementation ledger retains four non-scientific workflow events. None changed a scientific result file, allocation, estimator, threshold, or inference; the full trace remains available in the companion materials.

13.8 What the repaired design does and does not identify

The fully crossed design study establishes four bounded results.

1. The multilevel biological-design study tool/promoter alias can be removed prospectively while preserving animal, session, laboratory, memory-age, intervention-time, specificity, and footprint structure.
2. The resulting allocation has positivity and full column rank for the declared synthetic estimands.
3. Two label-free adapters and two capacity-matched bases can still disagree or fail under the same target, so method choice remains part of the result.
4. Power and set precision can remain inadequate after structural identifiability is repaired.

It does not identify a biological astrocyte as content, index, gate, eligibility trace, stabilizer, or

global state. It does not show that a promoter is cell-type specific, that a tool has a particular footprint, that a remembered item resides in an astrocyte, or that SAN's receiver-relative phase extension has been measured. The simulator has no biological phase variable, no neuronal bypass, no matched biological rescue, and no opened animal endpoint.

13.9 Sealed future biological test

The future test remains `SEALED_NOT_OPENED`. Before any biological outcome is accessed, a protocol derived from this gate must declare: the biological preparation and circuit; intervention versions; all tool-by-promoter cells or a justified estimable fraction; cell-identity and footprint assays; recent/remote and encoding/recall timing; complete-animal blocking; sex and other biological strata; simultaneous neuronal, astrocytic, physiological, and behavioral measurements; an eight-contrast confirmatory family; pilot-based variance and intraclass-correlation inputs; neuronal bypass or matched rescue; target-information access; uncertainty-set precision; and a refusal rule when content, access, stabilization, or general-state explanations remain observationally equivalent.

13.10 What the crossed study establishes

The fully crossed design study closes the exact allocation defect that the multilevel biological-design study exposed. All nine tool-by-promoter cells occur at registered replication in all twelve laboratory-split blocks; specificity and footprint vary separately; the design matrix is full rank; two unlabeled adapters and an equal-capacity nonlinear comparator are executed; 36,288 held-out predictions replay exactly; and power planning shows why structural repair alone cannot guarantee a decisive biological experiment.

The most important result is again not a universal success. CORAL alignment harms accuracy in three laboratories, empirical-quantile alignment does not repair transport, the nonlinear comparator supplies no general advantage, all 224 factorial readbacks remain non-significant after their global family, and many plausible biological effect/ICC scenarios require substantially more than six animals per crossed cell. The fully crossed design is therefore a validated causal-design repair and an adverse sensitivity result, not a biological validation. The bounded formal certificate below checks its finite allocation and estimand propositions. A biological protocol should be frozen only after pilot-based variance and effect inputs exist.

14. Bounded formal verification and evidence compatibility

14.1 The ordinary problem: a proof can be correct and still prove the wrong thing

The fully crossed design study made several exact claims: three tools and three promoters form nine cells; the declared split counts imply 468 synthetic animals; seven conditions imply 36,288 held-out predictions; two adapters do not use target labels; the linear and nonlinear readouts fit the same number of coefficients; non-rejection is not equivalence; and the biological endpoint remains closed. Those claims are small enough to check mechanically. Yet placing them in a theorem prover creates a new danger. A valid proof of hand-entered constants does not prove that a CSV was

parsed correctly, that a Python program implements the definitions, that an animal experiment satisfies causal assumptions, or that an astrocyte contains memory content.

The useful operation is therefore not “prove the paper.” It is to isolate the finite consequences whose failure would break the design’s own accounting, check those consequences, and attach an explicit boundary to every connection between the formal model and the executable evidence. Only after that problem is visible is **bounded formalization** an earned name for this proof boundary.

The accepted Lean 4 file contains twenty-six theorem declarations [34]. It was checked through the shared compiler lease with exit code zero; all twenty-six declarations report no axiom dependencies, and the source contains no `sorry`, `admit`, or declared `axiom`. A failed first check and an intermediate check containing a `propext` dependency are preserved. Only the final all-theorem check is the accepted certificate.

14.2 Crossed allocation and count invariants

The finite formal model gives each tool and promoter one of three constructors. Their cell index is

$$I(t, p) = 3I_T(t) + I_P(p), \quad 0 \leq I(t, p) < 9. \quad (58)$$

Lean checks the three tool constructors, three promoter constructors, nine-cell count, and valid index bound for every constructor pair. With registered replication (`n_d=4`), (`n_c=3`), and (`n_e=6`), it also checks

$$9n_d = 36, \quad 9n_c = 27, \quad 9n_e = 54, \quad 4(36 + 27 + 54) = 468. \quad (59)$$

The sessions, roles, and condition constants then give

$$468 \times 4 \times 6 = 11,232, \quad 4 \times 54 \times 4 \times 6 \times 7 = 36,288, \quad 36,288 \neq 24,192. \quad (60)$$

These theorems independently close the arithmetic defect recorded by the additive erratum. They do not import the allocation CSV. The empirical checks that all twelve laboratory-split matrices contain the registered replication, have numerical rank eleven, and keep specificity and footprint sufficiently separated remain deterministic Python/CSV validations. Their compatibility edge is yellow rather than green because no verified parser or refinement theorem connects those rows to Lean.

14.3 Access profiles and fitted-capacity equality

Every executed condition receives four Boolean access fields: target-development features, target-development labels, target-calibration labels, and permission to refit the classifier. The formal profiles preserve the distinction

condition	X_T^{dev}	Y_T^{dev}	Y_T^{cal}	refit
source only	0	0	0	0
CORAL or quantile	1	0	0	0
labeled calibration	0	0	1	0
full refit	1	1	1	1

(61)

Lean proves that CORAL and empirical-quantile alignment are label-free under this definition, labeled calibration uses target calibration labels without refitting, and full refit is not label-free. It also checks the declared capacity equality

$$(d_{map}, d_{out}, k)_{linear} = (28, 6, 174) = (d_{map}, d_{out}, k)_{nonlinear}, \quad 28 \times 174 = 4,872. \quad (62)$$

This is equality of fitted dimensions and coefficient counts. It is not a theorem of functional equivalence, equal inductive bias, equal approximation power, or biological plausibility. The model-receipt-to-Lean edge remains yellow because the access fields and constants are manually mapped and hash-checked rather than extracted by a verified program.

14.4 Non-rejection, adverse history, and the unopened seal

The formal interpretation model contains an explicit nonzero estimate whose rejection flag is false. It therefore proves the counterexample

$$\neg [\forall d, (\text{reject}(d) = 0) \Rightarrow (\hat{\tau}(d) = 0)]. \quad (63)$$

This is a logical firebreak, not a statistical reanalysis. It explains why the 224 factorial non-rejections remain unresolved instead of becoming 224 zero effects. The p values, intervals, Holm procedure, and animal-block analysis remain executable results outside Lean.

The evidence-history object records 334 generator-family transport adverse rows, 64 multilevel-design adverse rows, seven multilevel-design limitations, 374 fully crossed-design adverse rows, and four workflow events. Adding the formal certificate is defined to preserve that object exactly:

$$H_{formal} = H_{crossed} = (334, 64, 7, 374, 4). \quad (64)$$

Lean checks both preservation and all five counts. It also checks that a `sealedNotOpened` state makes the abstract biological endpoint inaccessible. Hash continuity, filesystem custody, row content, and protocol enforcement remain external validation obligations. No endpoint value is represented in the theorem file.

14.5 Claim signature and compatibility boundaries

The machine-readable claim signature identifies what the certificate provides, what it requires, its objects and constants, its source anchors, its forbidden overclaims, and its open edges. Eleven compatibility edges separate three states:

1. **Green:** finite arithmetic and logic proved inside the declared Lean model.

2. **Yellow:** manually transcribed or validator-bound connections from contracts, Python, JSON, CSV, result hashes, and seal custody to that model.
3. **Outside or red:** biological astrocyte roles, SAN/PWD/NAPOT/consciousness, efficacy, safety, treatment, and clinical claims that the certificate does not authorize.

This boundary is part of the result. Calling the certificate “a formal proof of multicellular engrams” would be false even though every accepted theorem compiles. The exact certified statement is narrower: the declared finite design constants and interpretation rules have the listed consequences.

14.6 Losslessness and strongest-form reconstruction

The formal layer does not rerun a scientific generator, change a threshold, exclude an animal block, revise a p value, remove an adverse row, change a source claim, or open a biological endpoint. Exact hashes preserve the complete developmental record separately.

The strongest-form audit rechecks the complete argument rather than a compressed substitute. The paper does not argue simply that “astrocytes store memories.” Its actual program separates content, index, gate, eligibility, stabilization, precision, and general state; asks which present cellular variables contribute to reconstruction; tests whether those roles remain distinguishable under perturbation, rescue, laboratory shift, information-access limits, and measurement change; and refuses a role verdict when alternatives remain observationally equivalent. The formal layer preserves that architecture. It does not narrow the paper to classification accuracy or revise a simplified caricature.

The reciprocal scientific check is equally specific. Current primary studies establish bounded astrocytic participation in synaptic regulation, plasticity, recall, stabilization, and precision in particular preparations. They do not supply the complete role-separation result sought here. That evidentiary boundary is an uncompleted experimental mapping, not a conflict between the full SAN source architecture and the literature.

14.7 Nonduplication boundary

This paper owns a specific scientific object: a neuron-astrocyte role taxonomy plus a cross-perturbation, rescue, transport, allocation, and uncertainty program for distinguishing content, access, eligibility, stabilization, precision, and general-state explanations. The adjacent Virtual Engram program owns the general carrier-invariance and cross-domain reconstruction question. The Qualia Tomogram program owns reconstruction from multiple receiver-conditioned projections. The Gamma-Shannon-PWD program owns tonic expectation, phase departure, and conditional self-information. The mobile-immune/COT program owns typed body-brain signaling by immune cells and soluble factors.

Those interfaces can exchange hypotheses without becoming interchangeable papers. The bounded formalization audit does not absorb a general virtual-engram theorem, a projection-slice reconstruction theory, a gamma-consciousness argument, or an immune communication model. Conversely, removing the virtual-engram consequence from this manuscript would break its source-faithful explanation of why stable recall can survive changing physical realization. The correct disposition is a bounded interface, not a merger and not deletion.

Within this manuscript, the cumulative study sequence is intentional evidence history rather than a

set of independent publications. Earlier benchmark reversals, adverse adapters, ambiguity results, the multilevel biological-design study alias, the fully crossed design study repair, and the count erratum remain visible because a later success must not erase the condition that motivated it.

14.8 What the formal certificate establishes

The bounded formal certificate passes its losslessness, strongest-form, and nonduplication audit. The final proof leaf contains twenty-six accepted, axiom-free theorem declarations under Lean’s audit output. The claim-signature and compatibility files prevent those theorems from being promoted into a biological conclusion. All prior manuscripts and adverse records remain preserved, and the biological endpoint remains `SEALED_NOT_OPENED`.

The formal certificate therefore closes the finite design-integrity questions it was built to answer while leaving the biological experiment explicitly open.

15. Predictions and falsifiers

Prediction 1: role diversity

Different astrocyte ensembles will produce different causal profiles. Some will primarily alter access or motivation; others will alter stability, precision, or neuronal representation. A single universal “astrocyte memory” score will underperform the role-separated model.

Prediction 2: slower state complements fast neuronal reconstruction

Astrocyte variables integrated over seconds to days will improve prediction of stability and precision more consistently than moment-to-moment cue identity. This prediction is aligned with the multiday eligibility result [14] but remains open across regions and tasks.

Prediction 3: carrier flexibility

As neuronal ensemble overlap changes over time, a relation-aware neuronal–astrocytic model should preserve prediction of recall better than a fixed-address model when address change also removes the fixed decoder’s task-relevant projection. The independent dynamical benchmark shows why both clauses are needed: carrier cosine can fall to 0.087 while the fixed-address decoder remains perfect. The factorial envelope independently varies address overlap and task projection, producing fixed-decoder accuracy as low as 0.445 while relational alignment remains at least 0.997 in the registered generator. Because that recovery is built from a relation-preserving synthetic family, the biological prediction remains open. A failure of relational recovery after fixed projection is demonstrably lost would narrow the virtual-engram proposal.

Prediction 4: receiver-relative residual

After controlling ordinary phase, rate, calcium, arousal, and cell identity, receiver-relative phase/state deviation will improve held-out prediction of route selection or successful reinstatement. If it does

not, the PWD extension is not needed for this phenomenon.

Falsifiers

The multicellular reconstruction model is narrowed if selective astrocyte variables add neither predictive nor causal value beyond neuronal and global-state models across adequately powered tasks. The content hypothesis fails if cue-specific information disappears after hidden-neuronal-state controls or if astrocyte manipulation changes only general fear, arousal, or motivation. The virtual-engram hypothesis fails in a preparation if successful recall strictly requires a fixed cellular ensemble and no relation-level invariant survives carrier change. The SAN phase extension fails if receiver-relative terms do not improve preregistered prediction or intervention choice beyond simpler models.

16. Discussion

The emerging astrocyte literature changes the memory question from “neurons or glia?” to “which coordinated cellular state performs which operation?” Local astrocyte processes can detect neuronal activity and alter the environment of synaptic transmission. Learning can recruit astrocyte ensembles. Manipulating those ensembles can affect recall, motivated behavior, stability, precision, fear-state representation, and neuronal readout. The first frozen benchmark makes one methodological consequence concrete: when several roles share the same natural cue correlation, observation alone can manufacture content language, while selective perturbation and rescue can recover role only inside a supported measurement regime. The independent dynamical benchmark prevents that result from becoming a new slogan. When roles also leave separable natural dynamical signatures, observation can outperform causal endpoint classification. The factorial envelope maps both lessons across 128 regimes and shows that no registered model view is globally sufficient. The empirical question is which observables and interventions identify the role in the preparation at hand.

The factorial result also changes how a biological study should be powered and interpreted. High marginal accuracy in an easy regime is not enough. A convincing result should report simultaneous role-set coverage, precision, false-content risk, carrier invariance, intervention spread, and transport under plausible nuisance shifts. It should predefine multiplicity families and preserve rank reversals rather than selecting the model that wins on average. In this sense, the 308 at-or-below-chance cells and the zero confirmed interaction effects are not failed paper outcomes. They identify the boundary between a mechanistic role claim and an underdetermined behavioral correlation.

The source-faithful SAN contribution is a joined architecture. The 2017 public discussion already treated a neuron-only snapshot as incomplete, connected fast neural activity to slower chemical and glial state, and framed recognition as reconstruction across never-identical physical patterns. The 2022 sources added receiver/transmitter arrays, cellular learned state, phase-coincidence relations, and neuron–glia participation in a distributed rendering process. The present paper converts that architecture into an experiment that can fail.

The virtual-engram proposal is physical, not ethereal. “Virtual” means that the useful identity is a relation reconstructed by the current network rather than a permanent address containing an unchanged copy. A melody remains identifiable across instruments; a visual object remains identifiable across retinal images; a memory may remain usable across changing cellular realizations. The analogy earns scientific content only through measurable invariants, longitudinal carrier change,

and causal tests.

The PWD extension is similarly bounded. Ordinary neuroscience already studies phase, timing, oscillations, neuromodulation, and astrocyte state. The proposed addition is a receiver-relative expected relation whose deviation has a downstream consequence. Astrocyte state may change the expectation or gain. That idea matters only if it outperforms simpler descriptions and supplies a better intervention.

The decisive advance is therefore methodological: record neurons and astrocytes together; separate content from access, stability, precision, motivation, and affect; manipulate each ensemble independently and jointly; validate tool footprints; compare nested models on held-out data and independent generator families; require matched and mismatched rescue; and retain an unresolved state when perturbation spread, missing variables, or measurement shift defeats identification. Carrier drift must also be decomposed into address change, decoder projection, relational recoverability, and behavioral preservation. Under those conditions, the phrase “multicellular engram” becomes a testable architecture rather than a rhetorical expansion of the word memory.

17. Conclusion

Memory recall is produced by a present biological system. The current neuronal ensemble carries activity through a local environment shaped by astrocytes, neuromodulators, extracellular ions, metabolism, and prior plasticity. Recent experiments show that selected astrocyte states can affect memory-related outcomes. The remaining problem is to identify their role.

This paper provides a role-separated model, an experimental route, and three frozen executable demonstrations of why the separation matters. It defines content, index, gate, eligibility, stabilization, precision, synergy, carrier flexibility, and receiver-relative phase as distinct hypotheses. In one synthetic family, cross-perturbation recovers roles that natural observation cannot. In the independent dynamical family, observation is strongest nominally, causal endpoints transport better under several shifts, temporal features are weak, and carrier rotation does not by itself require relational recovery. In the factorial family, every model has extensive failure regimes, every pairwise ranking reverses, simultaneous coverage often requires broad role sets, and no two-factor interaction survives the frozen multiplicity correction. The resulting biological program can determine whether a memory is best understood as a fixed neuronal trace, a neuronally represented state conditioned by glia, or a reconstructable multicellular relation whose carriers and context change over time—without allowing any one synthetic success, average, or model family to decide the answer.

References

1. Di Castro, M. A., Chuquet, J., Liaudet, N., et al. Local Ca^{2+} detection and modulation of synaptic release by astrocytes. *Nature Neuroscience* 14, 1276–1284 (2011). <https://doi.org/10.1038/nn.2929>
2. Henneberger, C., Papouin, T., Oliet, S. H. R. & Rusakov, D. A. Long-term potentiation depends on release of D-serine from astrocytes. *Nature* 463, 232–236 (2010). <https://doi.org/10.1038/nature08673>
3. Tanaka, M., Shih, P.-Y., Gomi, H., et al. Astrocytic Ca^{2+} signals are required for the functional integrity of tripartite synapses. *Molecular Brain* 6, 6 (2013). <https://doi.org/10.1186/1756-6606-6-6>
4. Sibille, J., Pannasch, U. & Rouach, N. Astroglial potassium clearance contributes to short-term plasticity of synaptically evoked currents at the tripartite synapse. *Journal of Physiology* 592, 87–102 (2014).

<https://doi.org/10.1113/jphysiol.2013.261735>

5. Adamsky, A., Kol, A., Kreisel, T., et al. Astrocytic activation generates de novo neuronal potentiation and memory enhancement. *Cell* 174, 59–71.e14 (2018). <https://doi.org/10.1016/j.cell.2018.05.002>
6. Armbruster, M., Naskar, S., Garcia, J. P., et al. Neuronal activity drives pathway-specific depolarization of peripheral astrocyte processes. *Nature Neuroscience* 25, 607–616 (2022). <https://doi.org/10.1038/s41593-022-01049-x>
7. Roy, D. S., Park, Y.-G., Kim, M. E., et al. Brain-wide mapping reveals that engrams for a single memory are distributed across multiple brain regions. *Nature Communications* 13, 1799 (2022). <https://doi.org/10.1038/s41467-022-29384-4>
8. Marks, T. D. & Goard, M. J. Stimulus-dependent representational drift in primary visual cortex. *Nature Communications* 12, 5169 (2021). <https://doi.org/10.1038/s41467-021-25436-3>
9. Schoonover, C. E., Ohashi, S. N., Axel, R. & Fink, A. J. P. Representational drift in primary olfactory cortex. *Nature* 594, 541–546 (2021). <https://doi.org/10.1038/s41586-021-03628-7>
10. Rupprecht, P., Duss, S. N., Becker, D., et al. Centripetal integration of past events in hippocampal astrocytes regulated by locus coeruleus. *Nature Neuroscience* 27, 927–939 (2024). <https://doi.org/10.1038/s41593-024-01612-8>
11. Sun, W., Liu, Z., Jiang, X., et al. Spatial transcriptomics reveal neuron–astrocyte synergy in long-term memory. *Nature* 627, 374–381 (2024). <https://doi.org/10.1038/s41586-023-07011-6>
12. Williamson, M. R., Kwon, W., Woo, J., et al. Learning-associated astrocyte ensembles regulate memory recall. *Nature* 637, 478–486 (2025), published online 6 November 2024. <https://doi.org/10.1038/s41586-024-08170-w>
13. Serra, I., Martín-Montegudo, C., Sánchez Romero, J., et al. Astrocyte ensembles manipulated with AstroLight tune cue-motivated behavior. *Nature Neuroscience* 28, 616–626 (2025). <https://doi.org/10.1038/s41593-025-01870-0>
14. Dewa, K., et al. The astrocytic ensemble acts as a multiday trace to stabilize memory. *Nature* 648, 146–156 (2025). <https://doi.org/10.1038/s41586-025-09619-2>
15. Bukalo, O., O’Sullivan, R., Tanisumi, Y., et al. Astrocytes enable amygdala neural representations supporting memory. *Nature* (2026). <https://doi.org/10.1038/s41586-025-10068-0>
16. Cooper, M. L., Selles, M. C., Cammer, M., et al. Astrocytes connect specific brain regions through plastic networks. *Nature* 655, 183–191 (2026). <https://doi.org/10.1038/s41586-026-10426-6>
17. Blumberg, M. *Neural Lace Podcast 4* (public audio, May 5, 2017; transcript Git-fixed October 22, 2022). <https://github.com/v5ma/selfawarenetworks/blob/09066e4b572ba36e145b3a1e0f31706fcc9388d6/nlp4.md>
18. Blumberg, M. Notes on Neural Oscillatory Tomography and other topics, a0115z.md (public Git fixation, June 10, 2022). <https://github.com/v5ma/selfawarenetworks/blob/a052ab363ca6a2f9538dc5da9f498d5c24365fca/a0115z.md>
19. Blumberg, M. Cellular Oscillating Tomography notes, a0270z.md (public Git fixation, August 7, 2022). <https://github.com/v5ma/selfawarenetworks/blob/20ed00b02326d72445ad91f5cad2274c1463bfb8/a0270z.md>
20. Kim, H., Lim, J., Kim, J., et al. Astrocytic ankyrin-2 enables memory persistence in the mouse hippocampus. *Nature Communications* 17, 5730 (2026). <https://doi.org/10.1038/s41467-026-75009-5>
21. Pouget, C., Morier, F., Autore, L., et al. Deconstruction of a memory engram reveals distinct ensembles recruited at learning. *Nature Neuroscience* 29, 1157–1165 (2026). <https://doi.org/10.1038/s41593-026-02230-2>
22. Abbaspoor, S., Aljishi, A. & Hoffman, K. L. Experience reorganizes content-specific memory traces in macaques. *Nature Neuroscience* 29, 1976–1986 (2026). <https://doi.org/10.1038/s41593-026-02357-2>

23. Ding, T., Angelopoulos, A. N., Bates, S., Jordan, M. I. & Tibshirani, R. J. Class-conditional conformal prediction with many classes. arXiv:2306.09335v2 (2023). <https://doi.org/10.48550/arXiv.2306.09335>
24. Gibbs, I., Cherian, J. J. & Candès, E. J. Conformal prediction with conditional guarantees. arXiv:2305.12616v4 (2024). <https://doi.org/10.48550/arXiv.2305.12616>
25. Chernozhukov, V., Wüthrich, K. & Yinchu, Z. Exact and robust conformal inference methods for predictive machine learning with dependent data. *Proceedings of Machine Learning Research* 75, 732-749 (2018). <https://proceedings.mlr.press/v75/chernozhukov18a.html>
26. Kitamura, T., Ogawa, S. K., Roy, D. S., et al. Engrams and circuits crucial for systems consolidation of a memory. *Science* 356, 73-78 (2017). <https://doi.org/10.1126/science.aam6808>
27. Taschenberger, G., Tereshchenko, J. & Kügler, S. A MicroRNA124 target sequence restores astrocyte specificity of gfaABC1D-driven transgene expression in AAV-mediated gene transfer. *Molecular Therapy – Nucleic Acids* 8, 13-25 (2017). <https://doi.org/10.1016/j.omtn.2017.03.009>
28. Gleichman, A. J., Kawaguchi, R., Sofroniew, M. V. & Carmichael, S. T. A toolbox of astrocyte-specific, serotype-independent adeno-associated viral vectors using microRNA targeting sequences. *Nature Communications* 14, 7426 (2023). <https://doi.org/10.1038/s41467-023-42746-w>
29. Gomez, J. L., Bonaventura, J., Lesniak, W., et al. Chemogenetics revealed: DREADD occupancy and activation via converted clozapine. *Science* 357, 503-507 (2017). <https://doi.org/10.1126/science.aan2475>
30. MacLaren, D. A. A., Browne, R. W., Shaw, J. K., et al. Clozapine N-oxide administration produces behavioral effects in Long-Evans rats: implications for designing DREADD experiments. *eNeuro* 3, ENEURO.0219-16.2016 (2016). <https://doi.org/10.1523/ENEURO.0219-16.2016>
31. Fisher, R. A. *The Design of Experiments*. Oliver and Boyd, Edinburgh (1935). <https://archive.org/details/in.ernet.dli.2015.502684>
32. Hernán, M. A. & Robins, J. M. *Causal Inference: What If*. Chapman & Hall/CRC (2020; continuously updated author edition). <https://miguelhernan.org/whatifbook>
33. Holm, S. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics* 6, 65-70 (1979). <https://doi.org/10.2307/4615733>
34. de Moura, L. & Ullrich, S. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, Lecture Notes in Computer Science 12699, 625-635 (2021). https://doi.org/10.1007/978-3-030-79876-5_37